

Downside Variance Risk Premium

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First Draft: December 2014

This Draft: May 2017

Abstract

We propose a new decomposition of the variance risk premium (*VRP*) in terms of upside and downside *VRPs*. These components reflect market compensation for changes in good and bad uncertainties. Empirically, we establish that the downside *VRP* is the main component of the *VRP*. We find a positive and significant link between the downside *VRP* and the equity premium, and a negative but statistically insignificant link between the upside *VRP* and the equity premium. The opposite relationships between these two components and the equity premium explains the stronger link found between the downside *VRP* and the equity premium compared with the well-established relationship between *VRP* and the equity premium. A simple equilibrium consumption-based asset pricing model, fitted to the U.S. data, supports our decomposition.

Keywords: Risk-neutral volatility, Realized volatility, Downside and upside variance risk premium, Skewness risk premium

JEL Classification: C58, G12

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§We gratefully acknowledge financial support from the Bank of Canada, UQAM research funds, and the IFSID. We thank Federico Bandi (the editor) and an anonymous referee for many insightful comments that improved the paper. We also thank seminar participants at the Federal Reserve Board, Johns Hopkins Carey Business School, Manchester Business School, University of North Carolina - Chapel Hill, University of Massachusetts at Amherst (Isenberg), Midwest Econometric Group Meeting 2013, CFE 2013, SNDE 2014, CIREQ 2015, SoFiE 2015, the Econometric Society World Congress 2015, and FMA 2015. We are grateful for conversations with Diego Amaya, Torben Andersen, Sirio Aramonte, Nikos Artavanis, Geert Bekaert, Bo Chang, Peter Christoffersen, Eric Engstrom, Nicola Fusari, Maren Hansen, Jianjian Jin, Olga Kolokolova, Lei Lian, Hening Liu, Bruce Mizrahi, Yang-Ho Park, Benoit Perron, Roberto Renò, George Tauchen, and Alex Taylor. We thank James Pinnington for research assistance. We also thank Bryan Kelly and Seth Pruitt for sharing their cross-sectional book-to-market index data. The views expressed in this paper are those of the authors. No responsibility for them should be attributed to the Bank of Canada or the Federal Reserve Board. Remaining errors are ours.

1 Introduction

A fundamental relationship in asset pricing posits a positive relation between risk and asset returns; see Merton (1973). This relationship holds over long horizons, as shown empirically by Bandi and Perron (2008) and Jacquier and Okou (2014), and theoretically by Bandi, Perron, Tamoni, and Tebaldi (2016) through decomposing the predictive relationship at different time scales. In short horizons, however, measures of risk based on historical asset prices lead to inconclusive estimates of this relation.¹

Two recent strands of the asset pricing literature have had success in addressing these conflicting short-horizon results caused by using historical measures of risk in empirical exercises. In one line of research, several studies have documented the asymmetry of risk-return trade off in response to negative and positive realizations in the financial markets. Among them, studies by Bonomo, Garcia, Meddahi, and Tédongap (2011); Feunou, Jahan-Parvar, and Tédongap (2013); and Rossi and Timmermann (2015) are related to our work. In particular, Feunou, Jahan-Parvar, and Tédongap (2013) explicitly model the upside and downside volatilities (the risk borne by market participants if realized returns exceed or fall below a certain threshold), document the impact of asymmetries on risk-return trade off, and highlight the role of downside risk. They develop a methodology that delivers an empirically robust, positive relation between risk and returns by allowing a time-varying market price of risk and asymmetric return distributions. In a second line of research, studies such as Bakshi and Kapadia (2003); Carr and Wu (2009); and Bollerslev, Tauchen, and Zhou (2009) rely on the information in option prices to measure time-varying risk compensations in the data. In particular, Bollerslev, Tauchen, and Zhou (2009) (henceforth, BTZ) study the variance risk premium (VRP), defined as the difference between the risk-neutral and physical expectations of return variation. The VRP , as formalized and studied by BTZ, is a robust predictor of asset returns at maturities of 3 to 6 months. Because of its significant predictive power for short-term asset returns, the VRP is often viewed as reflecting investors' appraisals of changes in near-future volatility.

In this paper, we bridge the gap between these two strands of the literature. We propose a new decomposition of the VRP in terms of upside and downside variance risk premia (VRP^U

¹Among many studies, Brandt and Kang (2004) observe a negative relation between realized market risk and returns. Ghysels, Santa-Clara, and Valkanov (2005) and Ludvigson and Ng (2007) find a positive relation. On the other hand, Baillie and DeGennaro (1990) and Bollerslev and Zhou (2006) document mixed results.

and VRP^D , respectively).² Our proposed decomposition is motivated by a simple assumption: Investors like good uncertainty as it increases the potential of substantial gains, but dislike bad uncertainty, as it increases the likelihood of severe losses. We define “good uncertainty” and “bad uncertainty” as volatility associated with positive and negative stock market returns, respectively. Given that investors dislike bad uncertainty, they are willing to pay a premium (the VRP^D) to hedge against variation in bad uncertainty. Hence we expect the VRP^D to be generally positive-valued. Conversely, because investors like good uncertainty, they should be willing to pay a premium (VRP^U) to be exposed to variation in good uncertainty. Thus, we expect a mostly negative-valued VRP^U . Thus, the (total) VRP that sums these two components lumps together market participants’ (asymmetric) views about good and bad uncertainties. As a result, a positive (total) VRP reflects the investors’ willingness to pay more in order to hedge against changes in bad uncertainty than for exposure to variations in good uncertainty. Hence, focusing on the (total) VRP does not provide a clear view of the trade-off between good and bad uncertainties, as a small positive VRP quantity does not necessarily imply a lower level of risk and/or risk aversion. Rather, it is an indication of a smaller difference between what agents are willing to pay for downside variation hedging versus upside variation exposure.

Using a non-parametric framework, we measure VRP^D and VRP^U and confirm our assertion. We further investigate whether disentangling the VRP improves the equity premium predictability. We find a positive and significant link between the VRP^D and the equity premium, and a negative but statistically insignificant link between the VRP^U and the equity premium. The stronger empirical link between the VRP^D and the equity premium (compared to the well-established relationship between VRP and the equity premium) is accounted by these opposite-signed relationships between VRP^D and VRP^U and the equity premium.

Theoretically, we support our empirical findings with a simple consumption-based equilibrium asset pricing model, where the representative agent is endowed with Epstein and Zin (1989) preferences, and where the consumption growth process is affected by distinct upside and downside shocks. Our model shares some features with Bansal and Yaron (2004); BTZ; Segal, Shaliastovich,

²We define the downside (upside) variance as the realized variance of the stock market returns for negative (positive) returns. The downside (upside) variance risk premium is the difference between option-implied and realized downside (upside) variance. Decomposing variance in this way was pioneered by Barndorff-Nielsen, Kinnebrock, and Shephard (2010).

and Yaron (2015); and Bekaert and Engstrom (2015); among others. We fit this model to the U.S. data and show that its implications are consistent with the documented salient regularities.

1.1 Related literature

This paper is related to the mounting literature on the properties of the *VRP*, as discussed in earlier works by Bakshi and Kapadia (2003); Vilkov (2008); and Carr and Wu (2009), among others. Theoretical attempts to rationalize the observed dynamics of the *VRP* have led to both reduced-form and general equilibrium models in the literature. Within the reduced-form framework, two papers have made significant contributions. An early study by Todorov (2010) focuses on the temporal dependence of continuous versus discontinuous *VRP* components within a semiparametric stochastic volatility model. He documents that both components exhibit nontrivial dynamics driven by *ex ante* volatility changes over time, coupled with *unanticipated* extreme swings in the market. Recently, Bandi and Renò (2016) focused on the role of a particular source of skewness in stock returns: the co-jumps (the dependence between discontinuous changes in asset prices and contemporaneous, discontinuous changes in volatility). The authors find that the co-jumps modify the mean of the return distribution while also inducing a *VRP*.

In a general equilibrium setting, BTZ design a simple model where time-varying volatility-of-volatility of consumption growth is the key determinant of the *VRP*. Drechsler and Yaron (2011) provide an equilibrium specification that features long-run risks and discontinuities in the stochastic volatility process governing the level of uncertainty about the cash flows. They extend the model of Bansal and Yaron (2004) by introducing a compound Poisson jump process in the state variable specification, thus departing from BTZ's assumption of Gaussian economic shocks. Our theoretical framework also extends BTZ's model, as we specify asymmetric predictable consumption growth components and differences of centered Gamma shocks to fundamentals.

Another strand of the literature explores the explanatory ability of the *VRP*. Along the time-series dimension, BTZ, Drechsler and Yaron (2011), and Kelly and Jiang (2014), among others, show that the *VRP* can help forecast the temporal variation in the aggregate stock market returns with high (low) premia predicting high (low) future returns, especially on a within-the-year time-scale. Ang, Hodrick, Xing, and Zhang (2006), and Cremers, Halling, and Weinbaum (2015), among

others, find that the price of variance risk successfully explains a large set of expected stock returns in the cross-section of assets.

Drawing on the decomposition of the quadratic variation of stock returns in terms of continuous and discontinuous variation, Bollerslev and Todorov (2011) decompose the *VRP* in terms of the diffusive and jump risk compensations. The authors show that the contribution of the jump tail risk premium is sizable. Our work is based on the alternative decomposition of the quadratic variation proposed by Barndorff-Nielsen, Kinnebrock, and Shephard (2010). The authors decompose the realized variance in terms of upside and downside semi-variances obtained by summing high-frequency positive and negative squared returns, respectively. Other authors have used the same decomposition of the realized variance with a focus on either realized variance predictability (Patton and Sheppard, 2015) or on equity risk premium predictability (Guo, Wang, and Zhou, 2015). These studies focus exclusively on realized measures and do not use option prices to infer the risk neutral counterparts and deduce the corresponding premia. In comparison, our work clearly evaluates the premia associated with upside and downside semi-variances, both realized and risk-neutral. In an independent and concurrent study, Kilic and Shaliastovich (2015) consider an alternative decomposition of the variance premium into the components associated with *good* and *bad* events and provide an economic model that explains their empirical findings.

Other related studies aim at decomposing the variance of macroeconomic variables. Segal, Shaliastovich, and Yaron (2015) study the impact of changes in *good* versus *bad* uncertainty on aggregate consumption growth and asset values. These authors demonstrate that these different types of uncertainties have opposite effects, with *good* (*bad*) economic risk implying a rise (decline) in future wealth or consumption growth. They characterize the role of asymmetric uncertainties in the determination of the economic activity level. Similarly, we develop and estimate a consumption-based equilibrium asset pricing model to highlight the roles that upside and downside variances play in pricing a risky asset in an otherwise standard model.

The rest of the paper proceeds as follows. In Section 2, we present our decomposition of the *VRP* and the method for construction of risk-neutral and realized semi-variances. Section 3 details the data used in this study and the empirical construction of predictive variables used in our analysis. We present and discuss our main empirical results in Section 4. Specifically, we intuitively describe

the components of variance risk premia, discuss their predictive ability, and explore the robustness of our findings. In Section 5, we introduce and estimate a simple equilibrium consumption-based asset pricing model that supports our empirical results. We evaluate the predictive ability of the difference between the VRP^U and VRP^D (labeled the skewness risk premium) in Section 6. Section 7 concludes. Further results regarding the extraction of risk-neutral quantities from options and robustness analysis are contained in an online Appendix.

2 Decomposition of the variance risk premium

In what follows, we decompose equity price changes into positive and negative returns with respect to a suitably chosen threshold. We set the threshold, κ , to zero in this study. It can assume other values, given the questions to be answered. We sequentially build measures for upside and downside semi-variances and for skewness. When taken to data, these measures are constructed non-parametrically.

We posit that equity market indices such as the S&P 500, S , are defined over the physical probability space characterized by $(\Omega, \mathbb{P}, \mathcal{F})$, where $\{\mathcal{F}_t\}_{t=0}^\infty \in \mathcal{F}$ are progressive filters on $\mathcal{F}$. The risk-neutral probability measure $\mathbb{Q}$ is related to the physical measure $\mathbb{P}$ through Girsanov's change of measure relation $\frac{d\mathbb{Q}}{d\mathbb{P}}|_{\mathcal{F}_T} = Z_T, T < \infty$. At time t , we denote total equity returns as $R_t^e = (S_t + D_t - S_{t-1})/S_{t-1}$, where D_t is the dividend paid out in period $[t-1, t]$. In sufficiently high sampling frequencies, D_t is equal to zero. Then, we denote the log of prices by $s_t = \ln S_t$, log-returns by $r_t = s_t - s_{t-1}$, and excess log-returns by $r_t^e = r_t - r_t^f$, where r_t^f is the risk-free rate observed at time $t-1$. We obtain cumulative excess returns by summing one-period excess returns, $r_{t \rightarrow t+k}^e = \sum_{j=0}^k r_{t+j}^e$, where k is our prediction horizon.

We build the VRP components following the steps in BTZ as the difference between option-implied and realized variances. Alternatively, these two components could be viewed as variances under risk-neutral and physical measures, respectively. In our approach, this construction requires three distinct steps: building the upside and downside realized variances, computing their expectations under the physical measure, and then doing the same under the risk-neutral measure.

2.1 Construction of the realized variance components

Following Andersen et al. (2003, 2001), we construct the realized variance of returns on any given trading day t as $RV_t = \sum_{j=1}^{n_t} r_{j,t}^2$, where $r_{j,t}^2$ is the j^{th} intraday squared log-return and n_t is the number of intraday returns recorded on that day. We add the squared overnight log-return (the difference in log price between when the market opens at t and when it closes at $t-1$), and we scale the RV_t series to ensure that the sample average realized variance equals the sample variance of daily logarithmic returns. For a given threshold κ , we decompose the realized variance into upside and downside realized variances following Barndorff-Nielsen, Kinnebrock, and Shephard (2010):

$$RV_t^U(\kappa) = \sum_{j=1}^{n_t} r_{j,t}^2 \mathbb{I}_{[r_{j,t} > \kappa]}, \quad (1)$$

$$RV_t^D(\kappa) = \sum_{j=1}^{n_t} r_{j,t}^2 \mathbb{I}_{[r_{j,t} \leq \kappa]}. \quad (2)$$

We add the squared overnight “positive” log-return (exceeding the threshold κ) to the upside realized variance RV_t^U , and the squared overnight “negative” log-return (falling below the threshold κ) to the downside realized variance RV_t^D . As the daily realized variance sums the upside and the downside realized variances, we apply the same scale to the two components of the realized variance. Specifically, we multiply both components by the ratio of the sample variance of daily log-returns over the sample average of the (pre-scaled) realized variance.

For a given horizon h , we obtain the cumulative realized quantities by summing the one-day realized quantities over h periods:

$$\begin{aligned} RV_{t,h}^U(\kappa) &= \sum_{j=1}^h RV_{t+j}^U(\kappa), \\ RV_{t,h}^D(\kappa) &= \sum_{j=1}^h RV_{t+j}^D(\kappa), \\ RV_{t,h} &= \sum_{j=1}^h RV_{t+j}(\kappa). \end{aligned} \quad (3)$$

By construction, the cumulative realized variance adds up the cumulative realized upside and

downside variances:

$$RV_{t,h} \equiv RV_{t,h}^U(\kappa) + RV_{t,h}^D(\kappa). \quad (4)$$

Barndorff-Nielsen, Kinnebrock, and Shephard (2010) laid down the theoretical underpinning of this decomposition by relying on a jump-diffusion process for the stock price

$$ds_t = \mu_t dt + \sigma_t dW_t + \Delta s_t,$$

where dW_t is an increment of the standard Brownian motion and $\Delta s_t \equiv s_t - s_{t-}$ refers to the jump component. The instantaneous variance can be defined as $\tilde{\sigma}_t^2 = \sigma_t^2 + (\Delta s_t)^2$. Under this general assumption on the instantaneous return process, and for $\kappa = 0$, the authors use infill asymptotics to demonstrate that

$$\begin{aligned} RV_{t,h}^U(0) &\xrightarrow{p} \frac{1}{2} \int_t^{t+h} \sigma_v^2 dv + \sum_{t \leq v \leq t+h} (\Delta s_v)^2 \mathbb{I}_{[\Delta s_v > 0]}, \\ RV_{t,h}^D(0) &\xrightarrow{p} \frac{1}{2} \int_t^{t+h} \sigma_v^2 dv + \sum_{t \leq v \leq t+h} (\Delta s_v)^2 \mathbb{I}_{[\Delta s_v \leq 0]}. \end{aligned}$$

2.2 Construction of the variance risk premium components

Next, we characterize the *VRP* of BTZ through premia accrued to bearing upside and downside variance risks, following these steps:

$$\begin{aligned} VRP_{t,h} &= \mathbb{E}_t^{\mathbb{Q}}[RV_{t,h}] - \mathbb{E}_t^{\mathbb{P}}[RV_{t,h}], \\ &= \left(\mathbb{E}_t^{\mathbb{Q}}[RV_{t,h}^U(\kappa)] - \mathbb{E}_t^{\mathbb{P}}[RV_{t,h}^U(\kappa)] \right) + \left(\mathbb{E}_t^{\mathbb{Q}}[RV_{t,h}^D(\kappa)] - \mathbb{E}_t^{\mathbb{P}}[RV_{t,h}^D(\kappa)] \right), \\ VRP_{t,h} &\equiv VRP_{t,h}^U(\kappa) + VRP_{t,h}^D(\kappa). \end{aligned} \quad (5)$$

Eq. (5) represents the decomposition of the *VRP* of BTZ into upside and downside variance risk premia – $VRP_{t,h}^U(\kappa)$ and $VRP_{t,h}^D(\kappa)$, respectively – that lies at the heart of our analysis.

2.2.1 Construction of $\mathbb{P}$ -expectation

The goal here is to evaluate $\mathbb{E}_t^{\mathbb{P}}[RV_{t,h}^U(\kappa)]$ and $\mathbb{E}_t^{\mathbb{P}}[RV_{t,h}^D(\kappa)]$. To this end, we consider three specifications: random walk (RW), upside/downside heteroscedastic autoregressive realized variance

(U/D-HAR), and multivariate heteroscedastic autoregressive realized variance (M-HAR).

Both U/D-HAR and M-HAR specifications mimic Corsi (2009)'s HAR-RV model. To get genuine expected values for realized measures that are not contaminated by forward bias or the use of contemporaneous data, we perform an out-of-sample forecasting exercise to predict the three realized variances, at different horizons, corresponding to 1, 2, 3, 6, 9, 12, 18, and 24 months ahead. We find that these alternative specifications provide qualitatively similar results, likely due to persistence in volatility. Hence, for simplicity and to save space, we only report the results based on the random walk model.

The random walk model is specified as

$$\mathbb{E}_t^{\mathbb{P}}[RV_{t,h}^{U/D}(\kappa)] = RV_{t-h,h}^{U/D}(\kappa),$$

where U/D stands for “ U or D ”. This is the model used in BTZ.³

2.2.2 Construction of $\mathbb{Q}$ -expectation

Let $F_t = S_t \exp(r_t^f h)$ denote the price of a future contract at time t , with maturity h . To build the risk-neutral expectation of $RV_{t,h}$, we follow the methodology of Andersen and Bondarenko (2007):

$$\begin{aligned} \mathbb{E}_t^{\mathbb{Q}}[RV_{t,h}^U(\kappa)] &\approx \mathbb{E}_t^{\mathbb{Q}}\left[\int_t^{t+h} \tilde{\sigma}_v^2 \mathbb{I}_{[\ln(F_v/F_t) > \kappa_F]} dv\right], \\ &= \mathbb{E}_t^{\mathbb{Q}}\left[\int_t^{t+h} \tilde{\sigma}_v^2 \mathbb{I}_{[F_v > F_t \exp(\kappa_F)]} dv\right], \end{aligned}$$

where κ_F is a threshold used to compute risk-neutral expectations of semi-variances.⁴ Thus,

$$\begin{aligned} \mathbb{E}_t^{\mathbb{Q}}[RV_{t,h}^U(\kappa)] &\approx 2 \int_{F_t \exp(\kappa_F)}^{\infty} \frac{M_0(\underline{S})}{\underline{S}^2} d\underline{S}, \\ M_0(\underline{S}) &= \min(P_t(\underline{S}), C_t(\underline{S})), \end{aligned} \tag{6}$$

³Specifications for U/D-HAR and M-HAR models are available from the authors upon request.

⁴Note that κ_F should be set to $(\kappa - r_t^f)h$ to get consistent thresholds when computing realized and option-implied quantities.

where, $P_t(\underline{S})$, $C_t(\underline{S})$, and $\underline{S}$ are prices of European put and call options (with maturity h), and the strike price of the underlying asset, respectively. Similarly for $\mathbb{E}_t^{\mathbb{Q}}[RV_{t,h}^D(\kappa)]$, we get

$$\mathbb{E}_t^{\mathbb{Q}}[RV_{t,h}^D(\kappa)] \approx 2 \int_0^{F_t \exp(\kappa_F)} \frac{M_0(\underline{S})}{\underline{S}^2} d\underline{S}. \quad (7)$$

Intuitively, one can view the above equalities as replicating option portfolios weighted across mon-eyness above (upside) and below (downside) a given threshold.

We simplify our notation by renaming $\mathbb{E}_t^{\mathbb{Q}}[RV_{t,h}^U(\kappa)]$ and $\mathbb{E}_t^{\mathbb{Q}}[RV_{t,h}^D(\kappa)]$ as

$$IV_{t,h}^U = \mathbb{E}_t^{\mathbb{Q}}[RV_{t,h}^U(\kappa)], \quad (8)$$

$$IV_{t,h}^D = \mathbb{E}_t^{\mathbb{Q}}[RV_{t,h}^D(\kappa)]. \quad (9)$$

We refer to $IV_{t,h}^{U/D}$ as the “risk-neutral semi-variance” or “implied semi-variance” of returns. These quantities are conditioned on the threshold value κ , which we suppress to simplify notation. As evident in this section, our measures of realized and implied volatility are model-free.

A number of recent studies have raised concerns about the accuracy of traditional methods for evaluating risk-neutral expectations of realized higher moments. For instance, papers by Orlowski (2017) and Schneider and Trojani (2015) document that a non-trivial fraction of information appears to be lost at short maturities (typically one week), as the traditional approach uses a feasible incomplete-market replicating option portfolio with a discrete set of options to approximate a complete market form, which requires a compact set of options.⁵ In our study, we focus on maturities of one month or longer. In an online Appendix, we document that both approaches yield similar (highly correlated, with correlation above 0.95) risk-neutral times-series dynamics at a monthly horizon and beyond. Thus, we expect our construction of upside and downside VRP components (at a one-month horizon and beyond) as well as their links to the equity premium to be robust to the incomplete-market discretization error.

⁵This is due to the fact that a continuum of traded options is not available in the market. The feasible incomplete market approach in the traditional risk-neutral expectation assessment may imply a sizable discretization error. Instead, the authors argue that $\mathbb{Q}$ -expectations should be obtained from the prices of dynamically hedged trading strategies that exactly deliver the realized moment (typically, realized variance and its up/down components) as a payoff. This alternative technology dynamically optimizes the option portfolio weights by minimizing the discrepancy between the hedged quantity and the received payoff.

3 Data

In this study, we adapt BTZ's methodology and use modified measures introduced in Section 2.2. We thus need suitable data to construct excess returns, realized semi-variances ($RV^{U/D}$), and risk-neutral semi-variances ($IV^{U/D}$). Throughout the study, we set $\kappa = 0$.

3.1 Excess returns

We first compute the excess returns by subtracting three-month U.S. Treasury bill rates from log-differences in the S&P 500 composite index, sampled at the end of each month. We downloaded monthly S&P 500 index returns and three-month T-Bill rates from the CRSP data base. As our study requires reliable high-frequency data and option-implied volatilities, our sample runs from September 1996 to March 2015. Panel A of Table 1 reports the descriptive statistics of monthly S&P 500 excess return series.

3.2 High-frequency data and realized variance components

We then use intraday S&P 500 data downloaded from the Institute of Financial Markets, to construct the daily $RV^{U/D}$ series. We sum the five-minute squared negative returns for the downside realized variance (RV^D) and the five-minute squared positive returns for the upside realized variance (RV^U). We next add the daily squared overnight negative returns to the downside semi-variance, and the daily squared overnight positive returns to the upside realized variance. The overnight returns are computed for 4:00 p.m. to 9:30 a.m. The total realized variance is obtained by adding the downside and the upside realized variances.⁶ To construct physical expectations of volatility measures, we use a random walk model to forecast the three realized variances at horizons ranging between 1 and 24 months ahead.

3.3 Options data and risk-neutral variances

We extract risk-neutral quantities from daily data of European-style put and call options on the S&P 500 index, available through the OptionMetrics Ivy database. To obtain consistent risk-neutral moments, we preprocess the data by applying the same filters as in Chang, Christoffersen,

⁶For the three series, we use a multiplicative scaling of the average total realized variance series to match the unconditional variance of S&P 500 returns. Hansen and Lunde (2006) discuss various approaches to adjusting open-to-close RV s.

and Jacobs (2013). Risk-neutral upside and downside variances ($IV^{U/D}$) are constructed using the model-free “corridor” implied volatility methodology discussed in Andersen, Bondarenko, and Gonzalez-Perez (2015); Andersen and Bondarenko (2007); and Carr and Madan (1999), among others.⁷

Panel B of Table 1 shows that risk-neutral volatility measures are persistent – $AR(1)$ parameters are all above 0.60 – and demonstrate significant skewness and excess kurtosis. It is also clear that the main factor behind volatility behavior is the downside variance. The contribution of upside volatility to risk-neutral volatility is considerably less than that of downside volatility.

3.4 Data for structural estimation

In Section 5.5, we fit the theoretical model developed throughout Section 5 to the U.S. macroeconomic and financial data. We use data sampled at a monthly frequency. Excess returns, realized and risk-neutral volatility components, and (upside and downside) variance risk premia are constructed as described earlier in this section. We use seasonally adjusted, monthly, per capita consumption growth, deflated based on chained 2009 prices and downloaded from the Federal Reserve Bank of St. Louis FRED II data bank, to compute log-growth rates of consumption. This series spans January 1999 to March 2015. Our measure for the risk-free rate is the three-month U.S. Treasury Bill rate. To construct the real risk-free rate, we regress the ex-post real three-month Treasury Bill yield on the nominal rate and past annualized inflation, available from the WRDS Treasury and Inflation database. The fitted values from this regression are the proxy for the ex-ante real interest rates. Panel E of Table 1 reports the descriptive statistics of consumption growth data.

4 Empirical Results

In this section, we provide economic intuition and empirical support for our proposed decomposition of the VRP . First, we describe the intuitively expected behavior of the components of the VRP , as well as some salient features of the size and variability of these components. As our approach

⁷Our data set contains a large set of option contracts. Moreover, the sample features comparable numbers of out-of-the-money (OTM) put and call contracts (especially in longer-horizon maturities from 18 to 24 months) that enable a precise computation of risk-neutral semi-variances. A detailed description of options data is provided in an online Appendix.

is nonparametric, these facts stand as guidelines for realistic models (reduced-form and general equilibrium). Second, we provide an extensive investigation of the predictability of the equity premium based on the VRP and its components. We empirically demonstrate the contribution of downside risk premium and characterize the sources of VRP predictability documented by BTZ. Third, we provide a comprehensive robustness study.

4.1 Description of the variance risk premium components

We view the VRP as the premium that a market participant is willing to pay to hedge against variations in future realized volatilities. It is expected to be positive, as rationalized within equilibrium frameworks in Section 1.1. We confirm these findings by reporting the summary statistics for the VRP in Table 1. We also plot the time series of the VRP and its components in Figure 1. We present measures based on random walk and univariate U/D-HAR forecasts of realized volatility and its components under the physical measure. Constructions of quantities based on multivariate HAR (M-HAR) are virtually identical to those obtained from univariate HAR.⁸ To save space, and because the results obtained from either the random walk or HAR methods are qualitatively similar, we only report findings based on random walk forecasts of realized volatility. The series plotted in Figure 1 demonstrate that while HAR-based quantities are more volatile than time series based on the random walk, the difference is mainly due to the magnitude of fluctuations, but not the fluctuations themselves. This observation may explain the similarities in empirical findings. As expected, from 1996 to 2015, we can see that the VRP is positive most of the time and remains high in uncertain episodes. It is important to emphasize that the economics of the VRP may be richer than that induced by the volatility-of-volatility. Alternative approaches reveal additional VRP drivers such as co-jumps, as formalized in Bandi and Renò (2016). Using high-frequency data and infinitesimal cross-moments, the authors elicit both theoretically and empirically, how contemporaneous (anti-correlated) large discontinuities in the joint dynamics of return and volatility contribute to the time variation of the VRP .

We argued in Section 1 that we intuitively expect negative-valued VRP^U and positive-valued VRP^D . Table 1 clearly illustrates these intuitions, as the average VRP^U is about -2.60% . More-

⁸U/D-HAR and M-HAR forecasts of realized volatility and its components are based on the methodology of Corsi (2009).

over, Figure 1 confirms that VRP^U is usually negative through our sample period. The same table reports an average VRP^D of roughly 5.21%, and in Figure 1 we observe that VRP^D is usually positive. In Section 5, we show that under mild assumptions these expectations about VRP components are supported by our theoretical model.

Table 1 also reveals highly persistent, negatively skewed, and fat-tailed empirical distributions for (downside/upside) variance premia. Nonetheless, upside variance appears more left-skewed and leptokurtic compared with (total) variance and downside variance risk premia.

4.2 Predictability of the equity premium

BTZ derive a theoretical model where the VRP emerges as the main driver of time variation in the equity premium. They show both theoretically and empirically that a higher VRP predicts higher future excess returns. Because the VRP sums downside and upside variance risk premia, BTZ's framework entails imposing the same coefficient on both (upside and downside) components of the VRP when they are jointly included in a predictive regression of excess returns. However, such a constraint seems very restrictive given the asymmetric views of investors on good uncertainty (preference for upward variability) versus bad uncertainty (aversion to downward variability). Sections 4.1 and 5 document that both VRP^D and VRP^U have intrinsically different features.

Our results are based on a simple linear regression of k -steps-ahead cumulative S&P 500 excess returns on values of a set of predictors that include the VRP , VRP^U , and VRP^D . Following the results of Ang and Bekaert (2007), reported Student's t -statistics are based on heteroscedasticity and serial correlation consistent standard errors that explicitly take into account the overlap in the regressions, as advocated by Hodrick (1992). The model used for our analysis is simply

$$r_{t \rightarrow t+k}^e = \beta_0 + \beta_1 x_t(h) + \epsilon_{t \rightarrow t+k}, \quad (10)$$

where $r_{t \rightarrow t+k}^e$ is the cumulative excess returns between time t and $t+k$, $x_t(h)$ is one of the predictors discussed in Section 2.2 at time t , h is the construction horizon of $x_t(h)$, and $\epsilon_{t \rightarrow t+k}$ is a zero-mean error term. We focus our discussion on the significance of the estimated slope coefficients (β_1 s), determined by the robust Student's- t statistics. We report the predictive ability of regressions, measured by the corresponding adjusted R^2 s. For highly persistent predictor variables, the R^2 s for

the overlapping multi-period return regressions must be interpreted with caution, as noted by BTZ and Jacquier and Okou (2014), among others.⁹

We decompose the contribution of our predictors to show that (1) predictability results documented by BTZ are driven by the VRP^D and (2) predictability results are mainly driven by risk-neutral expectations – thus, risk-neutral measures contribute more than realized measures.

Our empirical findings, presented in Tables 2 to 5, support our claims. In Panel A of Table 2, we show that the two main regularities uncovered by BTZ, the hump-shaped increase in robust Student's- t statistics and adjusted R^2 s reaching their maximum at $k = 3$ (one quarter ahead), are present in the data. These effects, however, weaken as the aggregation horizon (h) increases from one month to three months or more; the predictability pattern weakens and then largely disappears for $h > 6$.

Panel B of Table 2 reports the predictability results based on using VRP^D as the predictor. We observe the hump-shaped pattern for Student's t -statistics and the adjusted R^2 s reaching their maximum between $k = 3$ or $k = 6$ months. Moreover, these results are more robust to the aggregation horizon of the predictor. We notice that, in contrast to the VRP results – where predictability is only present for monthly or quarterly constructed risk premia – the VRP^D results are largely robust to aggregation horizons; the slope parameters are statistically different from zero even for annually constructed downside variance risk premia ($h = 12$). Moreover, the VRP^D results yield higher adjusted R^2 s compared with the VRP regressions at similar prediction horizons, an observation that we interpret as the superior ability of the VRP^D to explain the variation in aggregate excess returns. Finally, we notice a gradual shift in prediction results from the familiar one-quarter-ahead peak of predictability documented by BTZ to 9-to-12-month-ahead peaks once we increase the aggregation horizon h . Based on these results, we infer that the VRP^D is the likely candidate to explain the predictive power of VRP , documented by BTZ.

Our results for predictability based on the VRP^U , reported in Panel C of Table 2, are weak. The hump-shaped pattern in both robust Student's t -statistics and in adjusted R^2 s, while present, is significantly weaker than the results reported by BTZ. Once we increase the aggregation horizon,

⁹We consider alternative measures of model assessment, such as Goyal and Welch (2008) out-of-sample root mean squared errors ($RMSE_{OOS}$) in our robustness study. A complete set of results is available upon request, but is not reported to save space.

h , these results are lost. We conclude that bearing upside variance risk does not appear to be an important contributor to the equity premium and, hence, is not a good predictor of this quantity. In addition, we interpret these findings as the VRP^U having a low contribution to overall VRP .

At this point, it is natural to inquire about including both VRP components in a predictive regression. We present the empirical evidence from this estimation in Panel A of Table 5. After inclusion of the VRP^U and VRP^D in the same regression, the statistical significance of the VRP^U 's slope parameters are broadly lost. We also notice a sign change in Student's t -statistics associated with the estimated slope parameters of the VRP^U and VRP^D . This observation is not surprising. As we show in our equilibrium model, and also intuitively, risk-averse investors like variability in positive outcomes of returns but dislike it in negative outcomes. Hence, in a joint regression, we expect the coefficient of VRP^D to be positive and that of VRP^U to be negative.

We claim that the patterns discussed earlier – and, hence, the predictive power of the VRP , and VRP^D – are rooted in expectations. That is, the driving force behind our results, as well as those of BTZ, is expected risk-neutral measures of the volatility components. To show the empirical findings supporting our claim, we run predictive regressions using equation (10). Instead of using the premia employed so far, we use realized and risk-neutral measures of variances, up- and downside variances, and skewness for x_t based on our discussions in Section 2, respectively.

Our empirical findings using risk-neutral volatility measures are available in Table 3. In Panel A, we report the results of running a predictive regression when the predictor is the risk-neutral variance obtained from direct application of the Andersen and Bondarenko (2007) method. It is clear that the estimated slope parameters are statistically different from zero for $k \geq 3$ at most construction horizons h . The reported adjusted R^2 s also imply that the predictive regressions have explanatory power for aggregate excess return variations at $k \geq 3$. The same patterns are discernible for risk-neutral downside and upside variances (Panels B and C). Reported adjusted R^2 s are lower than those reported in Table 2, and these measures of variation yield statistically significant slope parameters at longer prediction horizons than what we observe for the VRP and its components. Taken together, these observations imply that using the premium (rather than the risk-neutral variation) yields better predictions.

However, in comparison with realized (physical) variation measures, risk-neutral measures yield

better results. The analysis using realized variation measures is available in Table 4. It is obvious that, by themselves, the realized measures do not yield reasonable predictability, an observation corroborated by the empirical findings of Bekaert, Engstrom, and Ermolov (2015). The majority of the estimated slope parameters are statistically indistinguishable from zero, and the adjusted R^2 s are low, especially at short horizons. Inclusion of both risk-neutral and realized variance components does not change our findings dramatically, as demonstrated in Panels B and C of Table 5.

Given the weak performance of realized measures, it is easy to conclude that realized variation plays a secondary role to risk-neutral variation measures in driving the predictability results documented by BTZ or in this study. However, we need both elements in the construction of the variance or skewness risk premia, as realized or risk-neutral measures individually possess inferior prediction power.

4.3 Robustness

We perform extensive robustness exercises to document the prediction power of the VRP^D for aggregate excess returns in the presence of traditional predictor variables. The goal is to highlight the contribution of our proposed variables in a wider empirical context. Simply put, we observe that the predictive power does not disappear when we include other pricing variables, implying that the VRP^D is not simply a proxy for other well-known pricing ratios.

Following BTZ and Feunou et al. (2014), among many others, we include equity pricing measures such as the log price–dividend ratio ($\log(p_t/d_t)$), lagged log price–dividend ratio ($\log(p_{t-1}/d_t)$), and log price–earnings ratio ($\log(p_t/e_t)$); yield and spread measures such as term spread (tms_t), the difference between 10-year U.S. Treasury Bond yields and 3-month U.S. Treasury Bill yields; default spread (dfs_t), defined as the difference between BBB and AAA corporate bond yields; CPI inflation ($infl_t$); the skewness risk premium (srp_t , introduced in Section 6); and, finally, Kelly and Pruitt (2013) partial least-squares-based, cross-sectional in-sample and out-of-sample predictors ($kpis_t$ and $kpos_t$, respectively).

We consider two periods for our analysis: our full sample – September 1996 to December 2010 – and a pre-Great Recession sample, September 1996 to December 2007. The latter ends at the

time as the BTZ sample. We report our empirical findings in Tables 6 to 9. These results are based on semi-annually aggregated excess returns and estimated for the one-month-ahead prediction horizon.¹⁰ In this robustness study, we scale the cumulative excess returns; we use $r_{t \rightarrow t+6}^e/6$ as the predicted value and regress it on a one-month lagged predictive variable.

Full-sample simple predictive regression results are available in Table 6. Among *VRP* components, only the downside *VRP* ($dvrp_t$) has slope parameters that are statistically different from zero and adjusted R^2 s comparable in magnitude with other pricing variables. Once we use $dvrp_t$ along with other pricing variables, we observe the following regularities in Table 7, which reports the joint multivariate regression results. First, the estimated slope parameter for $dvrp_t$ is statistically different from zero in all cases. Second, these regressions yield adjusted R^2 s that range between 3.10% (for $dvrp_t$ and tms_t , in line with findings of BTZ that report weak predictability for tms_t) and 25.71% (for $dvrp_t$ and $infl_t$).¹¹ The downside *VRP* in conjunction with the *VRP* or upside *VRP* remains statistically significant and yields adjusted R^2 s that are around 7%.

We obtain adjusted R^2 s that are lower than those reported by BTZ for quarterly and annually aggregated multivariate regressions. These differences are driven by the inclusion of the Great Recession period data in our full sample. To illustrate this point, we repeat our estimation with the data ending in December 2007. Simple predictive regression results based on these data are available in Table 8. Excluding the Great Recession period data improves even the univariate predictive regression adjusted R^2 s across the board. The estimated slope parameters are also closer to BTZ estimates and are generally statistically significant.

In Table 9, we report multivariate regression results, based on data from 1996 to 2007. We notice that once $dvrp_t$ is included in the regression model, the *VRP* and the upside *VRP* are no longer statistically significant. Other pricing variables – except for term spread, default spread, and inflation – yield slope parameters that are statistically significant. Thus, inflation seems to lack prediction power in this sub-sample. We do not observe statistically insignificant slope parameters

¹⁰A complete set of robustness checks, including monthly, quarterly, and annually aggregated excess returns results, are available in an online Appendix.

¹¹The dynamics of inflation during the Great Recession period mimic the behavior of our variance risk premia. Gilchrist et al. (2017) meticulously study the behavior of this variable in the period from 2007 to 2009. According to their study, both full and matched PPI inflation in their model display an aggregate drop in 2008 and 2009, while the reactions of financially sound and weak firms are asymmetric, with the former lowering prices and the latter raising prices in this period. Thus, the predictive power of this variable, given the inherent asymmetric responses, is not surprising.

for the downside variance risk premium except when we include vrp_t . Across the board, adjusted R^2 s are high in this sub-sample.

5 A simple equilibrium model

In this section, we present an equilibrium consumption-based asset pricing model that supports the proposed decomposition of the VRP in terms of upside and downside components. We estimate this model, using a maximum likelihood procedure. We view this exercise as one possible theoretical motivation for our empirical findings, and it is important to stress that this is not the only way. We rely on BTZ, but could have built on Bandi and Renò (2016) or Drechsler and Yaron (2011) as well. Our main objective is to highlight the roles that upside and downside variances play in pricing a risky asset in an otherwise standard asset pricing model. In particular, we show that the model, under standard and mild assumptions yields closed-form solutions for VRP components that align well with empirical regularities. In addition, fitting the model to the data yields empirical results that align well with model predictions, empirical regularities discussed in Section 4, and with BTZ findings. To save space, we only report the main results. Appendix A reports step-by-step derivations of the theoretical findings and the estimation procedure.

5.1 Preferences

We consider an endowment economy in discrete time. The representative agent's preferences over the future consumption stream are characterized by Kreps and Porteus (1978) intertemporal preferences, as formulated by Epstein and Zin (1989) and Weil (1989),

$$U_t = \left[(1 - \delta) C_t^{\frac{1-\gamma}{\theta}} + \delta \left(\mathbb{E}_t U_{t+1}^{1-\gamma} \right)^{\frac{1}{\theta}} \right]^{\frac{\theta}{1-\gamma}}, \quad (11)$$

where C_t is the consumption bundle at time t , δ is the subjective discount factor, γ is the coefficient of risk aversion, and ψ is the elasticity of intertemporal substitution (IES).¹² Parameter θ is defined

¹²Our theoretical findings do not depend on our choice of preferences. We use EZ in order to make our results comparable with BTZ; Segal, Shaliastovich, and Yaron (2015); or other similar papers. We have not attempted this exercise, but one can follow Bekaert and Engstrom (2015) and use habit-based preferences to obtain very similar theoretical results. The only additional ingredient needed in the Bekaert and Engstrom “BE-GE” set up is time-varying dynamics for both volatility-of-volatility components.

as $\theta \equiv \frac{1-\gamma}{1-\frac{1}{\psi}}$. If $\theta = 1$, then $\gamma = 1/\psi$ and Kreps and Porteus preferences collapse to the expected power utility, which implies an agent who is indifferent to the timing of the resolution of the consumption path uncertainty. With $\gamma > 1/\psi$, the agent prefers an early resolution of uncertainty. For $\gamma < 1/\psi$, the agent prefers a late resolution of uncertainty. Epstein and Zin (1989) show that the logarithm of the stochastic discount factor (SDF) implied by these preferences is given by

$$\ln M_{t+1} = m_{t+1} = \theta \ln \delta - \frac{\theta}{\psi} \Delta c_{t+1} + (\theta - 1) r_{c,t+1}, \quad (12)$$

where $\Delta c_{t+1} = \ln \left(\frac{C_{t+1}}{C_t} \right)$ is the log growth rate of aggregate consumption, and $r_{c,t}$ is the log return of the asset that delivers aggregate consumption as dividends. This asset represents the returns on a wealth portfolio. The Euler equation states that

$$\mathbb{E}_t [\exp(m_{t+1} + r_{i,t+1})] = 1, \quad (13)$$

where $r_{c,t}$ represents the log returns for the consumption-generating asset. The risk-free rate, which represents the returns on an asset that delivers a unit of consumption in the next period with certainty, is defined as

$$r_t^f = \ln \left[\frac{1}{\mathbb{E}_t(M_{t+1})} \right]. \quad (14)$$

5.2 Consumption dynamics under the physical measure

Our specification of consumption dynamics incorporates elements from Bansal and Yaron (2004); BTZ; Feunou, Jahan-Parvar, and Tédongap (2013); Segal, Shaliastovich, and Yaron (2015); and Bekaert and Engstrom (2015).

Fundamentally, we follow Bansal and Yaron (2004) in assuming that consumption growth has a predictable component. We differ from Bansal and Yaron in assuming that the predictable component is proportional to consumption growth's upside and downside volatility components. Thus, we are closer to Segal, Shaliastovich, and Yaron (2015).¹³ As a result, we have

$$\Delta c_{t+1} = \mu_0 + \mu_1 V_{u,t} + \mu_2 V_{d,t} + \sigma_c (\varepsilon_{u,t+1} - \varepsilon_{d,t+1}), \quad (15)$$

¹³Feunou, Jahan-Parvar, and Tédongap (2013), and Bekaert and Engstrom (2015) consider variations of this assumption in their studies.

where $\mu_1, \mu_2 \in \mathbb{R}$, $\varepsilon_{u,t+1}$ and $\varepsilon_{d,t+1}$ are two mean-zero shocks that affect both the realized and expected consumption growth.¹⁴ $\varepsilon_{u,t+1}$ represents upside shocks to consumption growth, and $\varepsilon_{d,t+1}$ denotes downside shocks. Following Bekaert and Engstrom (2015) and Segal, Shaliastovich, and Yaron (2015), we assume that these shocks have a demeaned Gamma distribution and model them as

$$\varepsilon_{i,t+1} = \tilde{\varepsilon}_{i,t+1} - V_{i,t}, \quad i = \{u, d\}, \quad (16)$$

where $\tilde{\varepsilon}_{i,t+1} \sim \Gamma(V_{i,t}, 1)$.¹⁵ These distributional assumptions imply that volatilities of upside and downside shocks are time varying and driven by shape parameters $V_{u,t}$ and $V_{d,t}$. In particular, we have

$$\text{Var}_t[\varepsilon_{i,t+1}] = V_{i,t}, \quad i = \{u, d\}. \quad (17)$$

Naturally, the total conditional variance of consumption growth when $\varepsilon_{u,t+1}$ and $\varepsilon_{d,t+1}$ are conditionally independent is simply $\sigma_c^2(V_{u,t} + V_{d,t})$.

The distribution of the shock to return shares similar properties with the difference of the demeaned Gamma in equation (15): (i) the variance is the sum of the variances of upside and downside shocks, and (ii) the skewness is (up to a scaling factor) the difference between the variances of upside and downside shocks.

The sign and size of μ_1 and μ_2 matter in this context. With $\mu_1 = \mu_2$, we have a stochastic volatility component in the conditional mean of the consumption growth process, similar to the classic GARCH-in-Mean structure for modeling risk-return trade-off in equity returns. With both slope parameters equal to zero, the model yields the BTZ unpredictable consumption growth.¹⁶ If $|\mu_1| = |\mu_2|$, with $\mu_1 > 0$ and $\mu_2 < 0$, we have Skewness-in-Mean, similar in spirit to the Feunou, Jahan-Parvar, and Tédongap (2013) formulation for equity returns. With $\mu_1 \neq \mu_2$, we have free parameters that have an impact on loadings of risk factors on risky asset returns and the stochastic

¹⁴This assumption is for the sake of brevity. Violating this assumption adds to algebraic complexity but does not affect our analytical findings.

¹⁵We use demeaned Gamma distributions is for the sake of tractability. One can use a number of alternative distributions with positive support and fat tails. For example, one may choose from compound Poisson, χ^2 , inverse Gaussian, or Lévy distributions.

¹⁶A consumption-based asset pricing model with a representative agent endowed with Epstein and Zin (1989) preferences and an unpredictable consumption growth process does not support the existence of distinct upside and downside variance risk premia with the expected signs. In particular, we have found that such a setting always yields a positive upside *VRP*.

discount factor. Intuitively, we expect $\mu_1 > 0$: A rise in upside volatility at time t implies higher consumption growth at time $t + 1$, all else being equal. By the same logic, we intuitively expect a negative-valued μ_2 , implying an expected fall in consumption growth following an uptick in downside volatility – following bad economic outcomes, households curb their consumption.

We observe that

$$\ln \mathbb{E}_t \exp(\nu \varepsilon_{i,t+1}) = f(\nu) V_{i,t}, \quad (18)$$

where $f(\nu) = -(\ln(1 - \nu) + \nu)$. Both Bekaert, Engstrom, and Ermolov (2015) and Segal, Shaliastovich, and Yaron (2015) use this compact functional form for the Gamma distribution cumulant. It simply follows that $f(\nu) > 0$, $f''(\nu) > 0$, and $f(\nu) > f(-\nu)$ for all $\nu > 0$.

We assume that $V_{i,t}$ follows a time-varying, square root process with time-varying volatility-of-volatility, similar to the specification of the volatility process in BTZ:

$$V_{u,t+1} = \alpha_u + \beta_u V_{u,t} + \sqrt{q_{u,t}} z_{t+1}^u, \quad (19)$$

$$q_{u,t+1} = \gamma_{u,0} + \gamma_{u,1} q_{u,t} + \varphi_u \sqrt{q_{u,t}} z_{t+1}^1, \quad (20)$$

$$V_{d,t+1} = \alpha_d + \beta_d V_{d,t} + \sqrt{q_{d,t}} z_{t+1}^d, \quad (21)$$

$$q_{d,t+1} = \gamma_{d,0} + \gamma_{d,1} q_{d,t} + \varphi_d \sqrt{q_{d,t}} z_{t+1}^2, \quad (22)$$

where z_t^i are standard normal innovations, and $i = \{u, d, 1, 2\}$. The parameters must satisfy the following restrictions: $\alpha_u > 0, \alpha_d > 0, \gamma_{u,0} > 0, \gamma_{d,0} > 0, |\beta_u| < 1, |\beta_d| < 1, |\gamma_{u,1}| < 1, |\gamma_{d,1}| < 1, \varphi_u > 0$, and $\varphi_d > 0$. In addition, we assume that $\{z_t^u\}, \{z_t^d\}, \{z_t^1\}$, and $\{z_t^2\}$ are *i.i.d.* $\sim N(0, 1)$ and jointly independent from $\{\varepsilon_{u,t}\}$ and $\{\varepsilon_{d,t}\}$.

The assumptions above yield time-varying uncertainty and asymmetry in consumption growth. Through volatility-of-volatility processes $q_{u,t}$ and $q_{d,t}$, the setup induces additional temporal variation in consumption growth. Temporal variation in the volatility-of-volatility process is necessary for generating a sizable *VRP*. Asymmetry is needed to generate upside and downside variance risk premia.

We solve the model following the methodology proposed by Bansal and Yaron (2004), BTZ, and many others. We consider that the logarithm of the wealth–consumption ratio w_t or the price–

consumption ratio ($pc_t = \ln\left(\frac{P_t}{C_t}\right)$) for the asset that pays the consumption endowment $\{C_{t+i}\}_{i=1}^{\infty}$ is affine with respect to state variables $V_{i,t}$ and $q_{i,t}$.

We then posit that the consumption-generating returns are approximately linear with respect to the log price-consumption ratio, as popularized by Campbell and Shiller (1988):

$$\begin{aligned} r_{c,t+1} &= \kappa_0 + \kappa_1 w_{t+1} - w_t + \Delta c_{t+1}, \\ w_t &= A_0 + A_1 V_{u,t} + A_2 V_{d,t} + A_3 q_{u,t} + A_4 q_{d,t}, \end{aligned}$$

where κ_0 and κ_1 are log-linearization coefficients, and A_0, A_1, A_2, A_3 , and A_4 are factor-loading coefficients to be determined. We solve for the consumption-generating asset returns, $r_{c,t}$, using the Euler equation (13). Following standard arguments (see Appendix A-1), we find the equilibrium values of coefficients A_0 to A_4 :

$$A_1 = -\frac{f\left[\sigma_c(1-\gamma)\right] + (1-\gamma)\mu_1}{\theta(\kappa_1\beta_u - 1)}, \quad (23)$$

$$A_2 = -\frac{f\left[-\sigma_c(1-\gamma)\right] + (1-\gamma)\mu_2}{\theta(\kappa_1\beta_d - 1)}, \quad (24)$$

$$A_3 = \frac{(1 - \kappa_1\gamma_{u,1}) - \sqrt{(1 - \kappa_1\gamma_{u,1})^2 - \theta^2\varphi_u^2\kappa_1^4 A_1^2}}{\theta\kappa_1^2\varphi_u^2}, \quad (25)$$

$$A_4 = \frac{(1 - \kappa_1\gamma_{d,1}) - \sqrt{(1 - \kappa_1\gamma_{d,1})^2 - \theta^2\varphi_d^2\kappa_1^4 A_2^2}}{\theta\kappa_1^2\varphi_d^2}, \quad (26)$$

$$A_0 = \frac{\ln \delta + \left(1 - \frac{1}{\psi}\right)\mu_0 + \kappa_0 + \kappa_1(\alpha_u A_1 + \alpha_d A_2 + \gamma_{u,0} A_3 + \gamma_{d,0} A_4)}{1 - \kappa_1}. \quad (27)$$

It is easy to see that while A_3 and A_4 are negative-valued, the signs of A_1 and A_2 depend on the signs and sizes of μ_1 and μ_2 . We report the conditions that ensure $A_1 > 0$ and $A_2 < 0$ after deriving the dynamic of the model under the risk-neutral measure.

5.3 Risk-neutral dynamics and the premia

Combining the historical dynamic and the stochastic discount factor imply the risk-neutral dynamic and closed-form expression for different risk premia (see Appendix A-2 for details on the

mathematical derivations). Starting with the equity risk premium, we have

$$\begin{aligned}
ERP_t &\equiv \mathbb{E}_t[r_{c,t+1}] - \mathbb{E}_t^{\mathbb{Q}}[r_{c,t+1}] \\
&= \frac{\gamma\sigma_c^2}{1+\gamma\sigma_c} V_{u,t} + \frac{\gamma\sigma_c^2}{1-\gamma\sigma_c} V_{d,t} \\
&\quad + (1-\theta)\kappa_1^2 (A_1^2 + A_3^2\varphi_u^2) q_{u,t} + (1-\theta)\kappa_1^2 (A_2^2 + A_4^2\varphi_d^2) q_{d,t}.
\end{aligned} \tag{28}$$

This expression for the equity risk premium shows that our model implies unequal loadings for upside and downside volatility factors. The slope coefficients for volatility-of-volatility factors are also, in general, unequal. To derive the upside and downside variance risk premia, we need to decompose the total variance of $r_{c,t+1}$. The total conditional variance ($\sigma_{r,t}^2 \equiv \text{Var}_t[r_{c,t+1}]$) is given by

$$\sigma_{r,t}^2 = \sigma_c^2 V_{u,t} + \sigma_c^2 V_{d,t} + \kappa_1^2 (A_1^2 + A_3^2\varphi_u^2) q_{u,t} + \kappa_1^2 (A_2^2 + A_4^2\varphi_d^2) q_{d,t}.$$

The upside and downside variances are

$$(\sigma_{r,t}^u)^2 = \sigma_c^2 V_{u,t} + \kappa_1^2 (A_1^2 + A_3^2\varphi_u^2) q_{u,t}, \tag{29}$$

$$(\sigma_{r,t}^d)^2 = \sigma_c^2 V_{d,t} + \kappa_1^2 (A_2^2 + A_4^2\varphi_d^2) q_{d,t}. \tag{30}$$

Hence, the upside and downside variance risk premia are given by

$$\begin{aligned}
VRP_t^U &\equiv \mathbb{E}_t^{\mathbb{Q}}[(\sigma_{r,t+1}^u)^2] - \mathbb{E}_t[(\sigma_{r,t+1}^u)^2] = (\theta-1)(\sigma_c^2\kappa_1 A_1 + \kappa_1^3 (A_1^2 + A_3^2\varphi_u^2) A_3\varphi_u^2) q_{u,t}, \\
VRP_t^D &\equiv \mathbb{E}_t^{\mathbb{Q}}[(\sigma_{r,t+1}^d)^2] - \mathbb{E}_t[(\sigma_{r,t+1}^d)^2] = (\theta-1)(\sigma_c^2\kappa_1 A_2 + \kappa_1^3 (A_2^2 + A_4^2\varphi_d^2) A_4\varphi_d^2) q_{d,t}.
\end{aligned}$$

As discussed before, we expect $VRP_t^U < 0$ and $VRP_t^D > 0$. It follows that

$$\sigma_c^2\kappa_1 A_1 + \kappa_1^3 (A_1^2 + A_3^2\varphi_u^2) A_3\varphi_u^2 > 0, \tag{31}$$

$$\sigma_c^2\kappa_1 A_2 + \kappa_1^3 (A_2^2 + A_4^2\varphi_d^2) A_4\varphi_d^2 < 0. \tag{32}$$

In the appendix A-3 we discuss the necessary and sufficient conditions ensuring that both inequalities (31) and (32) hold. We can express these conditions in a very simple and intuitive way:

- A sufficient condition for $VRP_t^d > 0$ is $\mu_2 \leq 0$,

- A necessary condition for $VRP_t^u < 0$ is $\mu_1 \geq \frac{f(\sigma_c(1-\gamma))}{\gamma-1} \geq 0$.

Our estimation delivers structural parameter estimates that satisfy all the theoretically expected restrictions – inducing $VRP_t^U < 0$ and $VRP_t^D > 0$ –, and are consistent with the facts discussed in depth in the first part of the paper. Since equation (43) implies that the equity risk premium loads positively on both $q_{u,t}$ and $q_{d,t}$, and because $VRP_t^U < 0$ is negatively proportional to $q_{u,t}$ while $VRP_t^D > 0$ is positively proportional to $q_{d,t}$, the equity risk premium loads positively on the downside VRP but negatively on the upside VRP . This feature of the general equilibrium model is also consistent with the empirical regularities documented in the predictability analysis.

5.4 Estimation

To appraise the empirical performance of our general equilibrium, we implement a maximum likelihood estimation procedure. Namely, we maximize the joint likelihood of consumption growth, stock market return, and risk-free rate series. The following steps provide a brief description of the estimation. The shocks to consumption growth and stock return are

$$\begin{aligned}\Delta c_{t+1} - \mathbb{E}_t[\Delta c_{t+1}] &= \epsilon_{1,t+1}, \\ r_{c,t+1} - \mathbb{E}_t(r_{c,t+1}) &= \epsilon_{1,t+1} + \epsilon_{2,t+1},\end{aligned}$$

where

$$\begin{aligned}\epsilon_{1,t+1} &\equiv \sigma_c(\varepsilon_{u,t+1} - \varepsilon_{d,t+1}), \\ \epsilon_{2,t+1} &\equiv \kappa_1 \left[(A_1 z_{t+1}^u + \varphi_u A_3 z_{t+1}^1) \sqrt{q_{u,t}} + (A_2 z_{t+1}^d + \varphi_d A_4 z_{t+1}^2) \sqrt{q_{d,t}} \right],\end{aligned}$$

and $\epsilon_{1,t+1}$, $\epsilon_{2,t+1}$ are conditionally independent random variables. Note that

$$\epsilon_{2,t+1} \sim N(0, \kappa_1^2 (A_1^2 + \varphi_u^2 A_3^2) q_{u,t} + \kappa_1^2 (A_2^2 + \varphi_d^2 A_4^2) q_{d,t}).$$

Hence, the joint density function of consumption and return writes

$$f_{(c,r_c)}(\Delta c_{t+1}, r_{c,t+1}) = f_{\epsilon_1}(\Delta c_{t+1} - \mathbb{E}_t[\Delta c_{t+1}]) f_{\epsilon_2}(r_{c,t+1} - \Delta c_{t+1} - (\mathbb{E}_t(r_{c,t+1}) - \mathbb{E}_t(r_{c,t+1}))),$$

where f_{ϵ_1} and f_{ϵ_2} are the marginal densities of $\epsilon_{1,t+1}$ and $\epsilon_{2,t+1}$, respectively. These marginal densities can be computed according to

$$f_{\epsilon_1}(\epsilon_{1,t+1}) = \frac{1}{\pi} \int_0^\infty \text{Re} [\exp(-i\nu\epsilon_{1,t+1} + f(i\sigma_c\nu)V_{u,t} + f(-i\sigma_c\nu)V_{d,t})] d\nu,$$

and

$$\begin{aligned} \ln[f_{\epsilon_2}(\epsilon_{2,t+1})] &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\kappa_1^2 (A_1^2 + \varphi_u^2 A_3^2) q_{u,t} + \kappa_1^2 (A_2^2 + \varphi_d^2 A_4^2) q_{d,t}) \\ &\quad - \frac{1}{2} \frac{\epsilon_{2,t+1}^2}{\kappa_1^2 (A_1^2 + \varphi_u^2 A_3^2) q_{u,t} + \kappa_1^2 (A_2^2 + \varphi_d^2 A_4^2) q_{d,t}}. \end{aligned}$$

It follows that the joint log-likelihood of consumption and return is calculated as

$$\ln L^{(C,R)} = \sum_{t=0}^{T-1} \ln(f_{(c,r_c)}(\Delta c_{t+1}, r_{c,t+1})), \quad (33)$$

where T denotes the sample size. The risk-free rate distribution is based on a Gaussian error likelihood

$$\ln L^{RF} \propto -\frac{1}{2} \sum_{t=1}^T \{\ln(RFRMSE^2) + e_t^2/RFRMSE^2\},$$

where the error term is computed as $e_t = r_{f_t}^{\text{observed}} - r_{f_t}^{\text{Model}}$, and the corresponding root-mean-square error is given by $RFRMSE \equiv \sqrt{\frac{1}{T} \sum_{t=1}^T e_t^2}$. Finally, all the parameters are estimated by maximizing the joint likelihood of consumption growth, stock return, and risk-free rate

$$\ln L^{(C,R)} + \ln L^{RF}. \quad (34)$$

To effectively implement our estimation strategy, we need $V_{u,t}$, $V_{d,t}$, $q_{u,t}$ and $q_{d,t}$ that are latent factors in the model. To circumvent this challenge, we assume that observed one-month ahead upside VRP (VRP_t^U), downside VRP (VRP_t^D), conditional upside stock return variance ($\mathbb{E}_t^{\mathbb{P}}[RV_{t+1}^U]$), and conditional downside stock return variance ($\mathbb{E}_t^{\mathbb{P}}[RV_{t+1}^D]$) are measured without error. This assumption entails that the observed quantities exactly match their theoretical counterparts, thus

allowing to infer the latent factors as

$$\begin{aligned}
q_{u,t} &= -\frac{1}{(1-\theta)\kappa_1(\sigma_c^2 A_1 + \kappa_1^2(A_1^2 + A_3^2 \varphi_u^2) A_3 \varphi_u^2)} V R P_t^U \equiv \varsigma_u V R P_t^U, \\
q_{d,t} &= -\frac{1}{(1-\theta)\kappa_1(\sigma_c^2 A_2 + \kappa_1^2(A_2^2 + A_4^2 \varphi_d^2) A_4 \varphi_d^2)} V R P_t^D \equiv \varsigma_d V R P_t^D, \\
V_{u,t} &= -\frac{\kappa_1^2}{\sigma_c^2} (A_1^2 + A_3^2 \varphi_u^2) \frac{\gamma_{u,0}}{\beta_u} - \frac{\alpha_u}{\beta_u} + \frac{1}{\beta_u \sigma_c^2} \mathbb{E}_t^{\mathbb{P}}[R V_{t+1}^U] - \frac{\kappa_1^2}{\sigma_c^2} (A_1^2 + A_3^2 \varphi_u^2) \frac{\gamma_{u,1}}{\beta_u} \varsigma_u V R P_t^U, \\
&\equiv \varpi_u + \vartheta_u \mathbb{E}_t^{\mathbb{P}}[R V_{t+1}^U] + \varrho_u V R P_t^U, \\
V_{d,t} &= -\frac{\kappa_1^2}{\sigma_c^2} (A_2^2 + A_4^2 \varphi_d^2) \frac{\gamma_{d,0}}{\beta_d} - \frac{\alpha_d}{\beta_d} + \frac{1}{\beta_d \sigma_c^2} \mathbb{E}_t^{\mathbb{P}}[R V_{t+1}^D] - \frac{\kappa_1^2}{\sigma_c^2} (A_2^2 + A_4^2 \varphi_d^2) \frac{\gamma_{d,1}}{\beta_d} \varsigma_d V R P_t^D, \\
&\equiv \varpi_d + \vartheta_d \mathbb{E}_t^{\mathbb{P}}[R V_{t+1}^D] + \varrho_d V R P_t^D.
\end{aligned}$$

Table (10) reports the structural parameter estimates of the general equilibrium model and their corresponding standard errors. These results clearly show that our general equilibrium model yields accurate parameter estimates that are consistent with the empirical evidence. Specifically, the estimated values confirm that $\mu_1 > 0$, $\mu_2 < 0$, and both parameters with different magnitudes ($|\mu_1| < |\mu_2|$) are economically and statistically significant.

5.5 Confronting the model with data

In implementing our structural estimation procedure, we show that model-implied upside equity returns variance, downside equity returns variance, and their corresponding premia perfectly match the observed series. Thus, there are at least three remaining empirical challenges for the equilibrium model: comparing structural model-implied to observed (1) predictive regression slopes of excess return on variance risk-premium components, (2) equity returns and consumption growth expectations, and (3) consumption growth variance dynamics.

Figure (2) graphically summarizes additional performance results of the equilibrium model. It shows that the predictive regression slopes as per equation (10) lie within the observed 95% confidence bounds. We also see that the equilibrium model-implied conditional variance of consumption growth tracks the EGARCH forecasts remarkably well. Overall, the proposed general equilibrium

model yields significant empirical support for its theoretical implications.

6 Skewness or signed jump risk premium

The difference between realized upside and downside variance is also known as the signed jump variation, see Patton and Sheppard (2015). The signed jump variation can be interpreted as a measure of (realized) skewness, see Feunou, Jahan-Parvar, and Tédongap (2013, 2016), since its expectation (under mild conditions) equals the conditional skewness.¹⁷ As a result, we use the terms “signed jump” and “realized skewness” interchangeably. The realized skewness between t and $t + h$ is computed as:

$$RSV_{t,h}(\kappa) = RV_{t,h}^U(\kappa) - RV_{t,h}^D(\kappa). \quad (35)$$

While the predictive ability of $RSV_{t,h}(0)$ has been studied in the literature (for example, by Guo, Wang, and Zhou, 2015), to be the best of our knowledge, this is the first study to investigate its premium. We define the skewness (or signed jump) risk premium (denoted by SRP) as the difference between the risk-neutral and physical expectations of the realized skewness (or signed jumps). It can be shown that this measure of the skewness risk premium is the spread between the upside and downside components of the VRP :

$$\begin{aligned} SRP_{t,h} &= \mathbb{E}_t^Q[RSV_{t,h}] - \mathbb{E}_t^P[RSV_{t,h}], \\ SRP_{t,h} &= VRP_{t,h}^U(\kappa) - VRP_{t,h}^D(\kappa). \end{aligned} \quad (36)$$

If $RSV_{t,h} < 0$, we view $SRP_{t,h}$ as a skewness premium – the compensation for an agent who bears downside risk. Alternatively, if $RSV_{t,h} > 0$, we view $SRP_{t,h}$ as a skewness discount – the amount that the agent is willing to pay to secure a positive return on an investment. This measure of the skewness risk premium is nonparametric and model-free. Table 1 reports that the average skewness risk premium is -7.8% . In Figure 1, we observe that SRP is generally negative-valued.

We study the predictive power of signed jump or skewness risk premium for aggregate returns by estimating Eq. (10). The equity premium prediction results using the SRP are reported in

¹⁷Refer to Appendix B for details.

Panel D of Table 2. It is clear that the *SRP* displays a stronger predictive power at longer horizons than the *VRP*. For monthly excess returns, the *SRP* slope coefficient is statistically different from zero at prediction horizons of 6 months ahead or longer. At $k = 6$, the adjusted R^2 of the *SRP* is comparable in size with that of the *VRP* (2.30% against 3.65%, respectively) and is strictly greater thereafter. At $k = 6$, the adjusted R^2 for the monthly excess return regression based on the *SRP* is smaller than that of the *VRP*^{*D*}. However, their sizes are comparable at $k = 9$ and $k = 12$ months ahead. Both trends strengthen as we consider higher aggregation levels for excess returns. At the semi-annual construction level ($h = 6$), the *SRP* already has more predictive power than both the *VRP* and *VRP*^{*D*} at a quarter-ahead prediction horizon. The increase in adjusted R^2 s of the *SRP* is not monotonic in the construction horizon level. We can detect a maximum at a roughly three-quarters-ahead prediction window for semi-annual and annually constructed *SRP*. This observation implies that the *SRP* is the intermediate link between one-quarter-ahead predictability using the *VRP* uncovered by BTZ and the long-term predictors such as the price–dividend ratio, dividend yield, or consumption–wealth ratio of Lettau and Ludvigson (2001). Given the generally unfavorable findings of Goyal and Welch (2008) regarding long-term predictors of equity premium, our findings regarding the predictive power of the *SRP* are particularly encouraging.

Panel D of Table 4 and Panel C of Table 5 report statistically significant slope parameters and notable adjusted R^2 s for realized skewness in long prediction horizons ($k \geq 6$) and for construction horizons ($h \geq 6$). Compared with the *SRP*, the realized skewness lacks predictive power in low construction or prediction horizons. Based on the results presented in Table 2, we argue that the *SRP* (and not the realized skewness) is a superior predictor, as it overcomes these shortcomings.

7 Conclusion

In this study, we have decomposed the variance risk premium – arguably one of the most successful short-term predictors of excess equity returns – to show that its prediction power stems from the downside *VRP* embedded in this measure. Market participants seem more concerned with market downturns and demand a premium for bearing that risk. By contrast, they seem to favor upward uncertainty in the market. We support this intuition through a simple equilibrium consumption-based asset pricing model able to replicate the main stylized facts observed in our

empirical investigation.

Empirically, the downside *VRP* demonstrates significant prediction power (that is at least as powerful as the *VRP*, and often stronger) for excess returns. We also show that the difference between upside and downside variance risk premia – the skewness risk premium – is a powerful predictor of excess returns. The skewness risk premium performs well for intermediate prediction steps beyond the reach of short-run predictors such as downside variance risk or variance risk premia and long-term predictors such as price–dividend or price–earnings ratios alike.

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Appendix

Appendix A: Details on equilibrium derivations

Appendix A-1: Deriving the log price-consumption ratio coefficients

We solve for the consumption-generating asset returns, $r_{c,t}$, using the Euler equation (13):

$$\mathbb{E}_t \left[\exp \left[\theta \ln \delta - \frac{\theta}{\psi} \Delta c_{t+1} + (\theta - 1)r_{c,t+1} + r_{c,t+1} \right] \right] = 1$$

and

$$\ln \mathbb{E}_t \left[\exp \left[\theta \ln \delta - \frac{\theta}{\psi} \Delta c_{t+1} + \theta r_{c,t+1} \right] \right] = 0.$$

Substituting for Δc_{t+1} , $r_{c,t+1}$, $V_{i,t+1}$ and $q_{i,t+1}$, we get:

$$\begin{aligned} \ln \mathbb{E}_t \left[\exp \left[\theta \ln \delta + (1 - \gamma) [\mu_0 + \mu_1 V_{u,t} + \mu_2 V_{d,t}] + \theta [\kappa_0 + (\kappa_1 - 1)A_0] + \theta (\alpha_u A_1 + \alpha_d A_2 + \gamma_{u,0} A_3 + \gamma_{d,0} A_4) \right. \right. \\ \left. \left. + (1 - \gamma) \sigma_c (\varepsilon_{u,t+1} - \varepsilon_{d,t+1}) \right. \right. \\ \left. \left. + \theta [A_1 (\kappa_1 \beta_u - 1) V_{u,t} + A_2 (\kappa_1 \beta_d - 1) V_{d,t} + A_3 (\kappa_1 \gamma_{u,1} - 1) q_{u,d} + A_4 (\kappa_1 \gamma_{d,1} - 1) q_{d,t}] \right. \right. \\ \left. \left. + \theta \kappa_1 \left(A_1 \sqrt{q_{u,t}} z_{t+1}^u + A_2 \sqrt{q_{d,t}} z_{t+1}^d + A_3 \varphi_u \sqrt{q_{u,t}} z_{t+1}^1 + A_4 \varphi_d \sqrt{q_{d,t}} z_{t+1}^d \right) \right] \right] = 0, \end{aligned} \quad (37)$$

and then proceed to compute the expectations and coefficients, as follows:

1. $\ln \mathbb{E}_t \left[\exp \left[\sigma_c (1 - \gamma) \varepsilon_{u,t+1} \right] \right] = f [\sigma_c (1 - \gamma)] V_{u,t},$
2. $\ln \mathbb{E}_t \left[\exp \left[-\sigma_c (1 - \gamma) \varepsilon_{d,t+1} \right] \right] = f [-\sigma_c (1 - \gamma)] V_{d,t}$
3. $\ln \mathbb{E}_t \left[\exp \left[\theta \kappa_1 (A_1 \sqrt{q_{u,t}} z_{t+1}^u + A_2 \sqrt{q_{d,t}} z_{t+1}^d + A_3 \varphi_u \sqrt{q_{u,t}} z_{t+1}^1 + A_4 \varphi_d \sqrt{q_{d,t}} z_{t+1}^d) \right] \right] =$
4. $\frac{1}{2} \theta^2 \kappa_1^2 \left[(A_1^2 + \varphi^2 A_3^2) q_{u,t} + (A_2^2 + \varphi^2 A_4^2) q_{d,t} \right],$
5. $\ln \mathbb{E}_t \left[\exp \left[(1 - \gamma) [\mu_1 V_{u,t} + \mu_2 V_{d,t}] + \theta [A_1 (\kappa_1 \beta_u - 1) V_{u,t} + A_2 (\kappa_1 \beta_d - 1) V_{d,t} + A_3 (\kappa_1 \gamma_{u,1} - 1) q_{u,d} + \right. \right. \right. \\ \left. \left. \left. A_4 (\kappa_1 \gamma_{d,1} - 1) q_{d,t} \right] \right] \right] =$
6. $(1 - \gamma) [\mu_1 V_{u,t} + \mu_2 V_{d,t}] + \theta [A_1 (\kappa_1 \beta_u - 1) V_{u,t} + A_2 (\kappa_1 \beta_d - 1) V_{d,t} + A_3 (\kappa_1 \gamma_{u,1} - 1) q_{u,d} + \\ A_4 (\kappa_1 \gamma_{d,1} - 1) q_{d,t}],$
7. $\ln \mathbb{E}_t \left[\exp \left[\theta \ln \delta + (1 - \gamma) \mu_0 + \theta [\kappa_0 + (\kappa_1 - 1)A_0] + \theta (\alpha_u A_1 + \alpha_d A_2 + \gamma_{u,0} A_3 + \gamma_{d,0} A_4) \right] \right] =$
8. $\theta \left[\ln \delta + \frac{1-\gamma}{\theta} \mu_0 + \kappa_0 + (\kappa_1 - 1)A_0 + \kappa_1 (\alpha_u A_1 + \alpha_d A_2 + \gamma_{u,0} A_3 + \gamma_{d,0} A_4) \right].$

We gather the terms for $V_{u,t}$, $V_{d,t}$, $q_{u,t}$ and $q_{d,t}$, and solve for A_0 to A_4 :

$$A_1 = -\frac{f\left[\sigma_c(1-\gamma)\right] + (1-\gamma)\mu_1}{\theta(\kappa_1\beta_u - 1)}, \quad (38)$$

$$A_2 = -\frac{f\left[-\sigma_c(1-\gamma)\right] + (1-\gamma)\mu_2}{\theta(\kappa_1\beta_d - 1)}, \quad (39)$$

$$A_3 = \frac{(1 - \kappa_1\gamma_{u,1}) - \sqrt{(1 - \kappa_1\gamma_{u,1})^2 - \theta^2\varphi_u^2\kappa_1^4A_1^2}}{\theta\kappa_1^2\varphi_u^2}, \quad (40)$$

$$A_4 = \frac{(1 - \kappa_1\gamma_{d,1}) - \sqrt{(1 - \kappa_1\gamma_{d,1})^2 - \theta^2\varphi_d^2\kappa_1^4A_2^2}}{\theta\kappa_1^2\varphi_d^2}, \quad (41)$$

$$A_0 = \frac{\ln \delta + \left(1 - \frac{1}{\psi}\right)\mu_0 + \kappa_0 + \kappa_1(\alpha_u A_1 + \alpha_d A_2 + \gamma_{u,0} A_3 + \gamma_{d,0} A_4)}{1 - \kappa_1}. \quad (42)$$

Appendix A-2: Analytical derivations of equity premium and variance risk premium components

We begin by deriving the risk-neutral distribution of all the shocks, $\varepsilon_{u,t+1}$, $\varepsilon_{d,t+1}$, z_{t+1}^u , z_{t+1}^d , z_{t+1}^1 and z_{t+1}^2 . In this computation, we construct the characteristic function for each shock and exploit the properties of characteristic functions to derive the expectations under the risk-neutral measure. Thus, our derivations yield exact equity and risk premia measures, in contrast to approximate values reported, for example, in equation (15) of Bollerslev, Tauchen, and Zhou (2009) or in Drechsler and Yaron (2011).

We start from $\varepsilon_{u,t+1}$. The SDF is the Radon-Nikodym change of measure and $\ln \mathbb{E}_t(M_{t+1})$ is the risk-neutral drift term. We have:

$$\begin{aligned} & \mathbb{E}_t^Q(\exp(\nu \varepsilon_{u,t+1})) \\ &= \mathbb{E}_t \left[\frac{M_{t+1}}{\mathbb{E}_t(M_{t+1})} \exp(\nu \varepsilon_{u,t+1}) \right] \\ &= \mathbb{E}_t [\exp(\nu \varepsilon_{u,t+1} + m_{t+1} - \ln(\mathbb{E}_t(M_{t+1})))] \\ &= \mathbb{E}_t \left[\exp \left(\nu \varepsilon_{u,t+1} + \theta \ln \delta - \frac{\theta}{\psi} \Delta c_{t+1} + (\theta - 1)r_{c,t+1} - \ln(\mathbb{E}_t(M_{t+1})) \right) \right] \\ &= \mathbb{E}_t \left[\exp \left(\begin{array}{c} \nu \varepsilon_{u,t+1} + \theta \ln \delta - \frac{\theta}{\psi} \Delta c_{t+1} \\ + (\theta - 1)(\kappa_0 + \kappa_1 w_{t+1} - w_t + \Delta c_{t+1}) - \ln(\mathbb{E}_t(M_{t+1})) \end{array} \right) \right] \\ &\equiv \mathbb{E}_t [\exp((\nu - \gamma \sigma_c) \varepsilon_{u,t+1} + \mathbb{B}_t^{*,1})] \end{aligned}$$

where

$$\mathbb{B}_t^{*,1} = \ln \left[\mathbb{E}_t \left[\exp \left(\begin{array}{c} \theta \ln \delta - \gamma(\mu_0 + \mu_1 V_{u,t} + \mu_2 V_{d,t} - \sigma_c \varepsilon_{d,t+1}) \\ + (\theta - 1)(\kappa_0 + \kappa_1 w_{t+1} - w_t) - \ln(\mathbb{E}_t(M_{t+1})) \end{array} \right) \right] \right].$$

Hence, it follows that:

$$\mathbb{E}_t^Q(\exp(\nu \varepsilon_{u,t+1})) = \exp(\mathbb{B}_t^{*,1} + f(\nu - \gamma \sigma_c) V_{u,t}).$$

With $\nu = 0$, we have

$$1 = \exp(\mathbb{B}_t^1 + f(-\gamma \sigma_c) V_{u,t}),$$

hence, we obtain:

$$\mathbb{B}_t^1 = -f(-\gamma \sigma_c) V_{u,t}.$$

In conclusion

$$\mathbb{E}_t^Q(\exp(\nu \varepsilon_{u,t+1})) = \exp((f(\nu - \gamma \sigma_c) - f(-\gamma \sigma_c)) V_{u,t})$$

Similarly for $\varepsilon_{d,t+1}$, we deduce that:

$$\mathbb{E}_t^{\mathbb{Q}}(\exp(\nu \varepsilon_{d,t+1})) = \exp((f(\nu + \gamma \sigma_c) - f(\gamma \sigma_c)) V_{d,t})$$

It follows that:

$$\begin{aligned}\mathbb{E}_t^{\mathbb{Q}}[\varepsilon_{u,t+1}] &= f'(-\gamma \sigma_c) V_{u,t} = -\frac{\gamma \sigma_c}{1 + \gamma \sigma_c} V_{u,t} \\ \mathbb{E}_t^{\mathbb{Q}}[\varepsilon_{d,t+1}] &= f'(\gamma \sigma_c) V_{d,t} = \frac{\gamma \sigma_c}{1 - \gamma \sigma_c} V_{d,t}\end{aligned}$$

We now derive the expectation of Gaussian shocks, $z_{t+1}^u, z_{t+1}^d, z_{t+1}^1$ and z_{t+1}^2 , under the risk-neutral measure, $\mathbb{Q}$. We start by deriving the characteristic function for z_{t+1}^u :

$$\begin{aligned}& \mathbb{E}_t^{\mathbb{Q}}(\exp(\nu z_{t+1}^u)) \\ &= \mathbb{E}_t \left[\frac{M_{t+1}}{\mathbb{E}_t(M_{t+1})} \exp(\nu z_{t+1}^u) \right] \\ &= \mathbb{E}_t \left[\exp(\nu z_{t+1}^u + m_{t+1} - \ln(\mathbb{E}_t(M_{t+1}))) \right] \\ &= \mathbb{E}_t \left[\exp \left(\nu z_{t+1}^u + \theta \ln \delta - \frac{\theta}{\psi} \Delta c_{t+1} + (\theta - 1) r_{c,t+1} - \ln(\mathbb{E}_t(M_{t+1})) \right) \right] \\ &= \mathbb{E}_t \left[\exp \left(\begin{aligned} & \nu z_{t+1}^u + \theta \ln \delta - \frac{\theta}{\psi} \Delta c_{t+1} \\ & + (\theta - 1)(\kappa_0 + \kappa_1 w_{t+1} - w_t + \Delta c_{t+1}) - \ln(\mathbb{E}_t(M_{t+1})) \end{aligned} \right) \right] \\ &\equiv \mathbb{E}_t \left[\exp \left((\nu + (\theta - 1)\kappa_1 A_1 \sqrt{q_{u,t}}) z_{t+1}^u + \mathbb{B}_t^{*,g1} \right) \right]\end{aligned}$$

where

$$\mathbb{B}_t^{*,g1} = \ln \left[\mathbb{E}_t \left[\exp \left(\begin{aligned} & (\theta - 1)\kappa_1 A_1 (\alpha_u + \beta_u V_{u,t}) + \theta \ln \delta - \gamma \Delta c_{t+1} \\ & + (\theta - 1)(\kappa_0 + \kappa_1 (w_{t+1} - A_1 V_{u,t+1}) - w_t) - \ln(\mathbb{E}_t(M_{t+1})) \end{aligned} \right) \right] \right].$$

It follows that:

$$\mathbb{E}_t^{\mathbb{Q}}(\exp(\nu z_{t+1}^u)) = \exp \left(\mathbb{B}_t^{*,g1} + \frac{(\nu + (\theta - 1)\kappa_1 A_1 \sqrt{q_{u,t}})^2}{2} \right).$$

Setting $\nu = 0$, we have

$$1 = \exp \left(\mathbb{B}_t^{g1} + \frac{((\theta - 1)\kappa_1 A_1 \sqrt{q_{u,t}})^2}{2} \right),$$

and hence:

$$\mathbb{B}_t^{g1} = -\frac{(\theta - 1)^2 \kappa_1^2 A_1^2 q_{u,t}}{2}.$$

Thus:

$$\begin{aligned}\mathbb{E}_t^{\mathbb{Q}}(\exp(\nu z_{t+1}^u)) &= \exp \left(\frac{(\nu + (\theta - 1)\kappa_1 A_1 \sqrt{q_{u,t}})^2 - ((\theta - 1)\kappa_1 A_1 \sqrt{q_{u,t}})^2}{2} \right) \\ &= \exp \left(\frac{\nu^2}{2} + \nu(\theta - 1)\kappa_1 A_1 \sqrt{q_{u,t}} \right).\end{aligned}$$

Based on the last result, we characterize the distribution of z_{t+1}^u under the risk-neutral measure as:

$$z_{t+1}^u \sim^{\mathbb{Q}} N((\theta - 1)\kappa_1 A_1 \sqrt{q_{u,t}}, 1)$$

Similarly, we characterize the distributions for the remaining shocks:

$$\begin{aligned} z_{t+1}^d &\sim \mathbb{Q}N((\theta - 1)\kappa_1 A_2 \sqrt{q_{d,t}}, 1) \\ z_{t+1}^1 &\sim \mathbb{Q}N((\theta - 1)\kappa_1 A_3 \varphi_u \sqrt{q_{u,t}}, 1) \\ z_{t+1}^2 &\sim \mathbb{Q}N((\theta - 1)\kappa_1 A_4 \varphi_d \sqrt{q_{d,t}}, 1) \end{aligned}$$

Thus far, we have derived the distribution of shock processes under the risk-neutral measure, $\mathbb{Q}$. Since any premium – whether equity, variance risk, or skewness risk premia – can be defined as the difference between the physical and risk-neutral expectations of processes, we are now ready to compute all the premia of interest. We start with the equity risk premium:

$$\begin{aligned} ERP_t &\equiv \mathbb{E}_t[r_{c,t+1}] - \mathbb{E}_t^\mathbb{Q}[r_{c,t+1}] \\ &= \mathbb{E}_t[\kappa_0 + \kappa_1 w_{t+1} - w_t + \Delta c_{t+1}] - \mathbb{E}_t^\mathbb{Q}[\kappa_0 + \kappa_1 w_{t+1} - w_t + \Delta c_{t+1}] \\ &= \kappa_1 \left(\mathbb{E}_t[w_{t+1}] - \mathbb{E}_t^\mathbb{Q}[w_{t+1}] \right) + \mathbb{E}_t[\Delta c_{t+1}] - \mathbb{E}_t^\mathbb{Q}[\Delta c_{t+1}] \end{aligned}$$

As it is clear from the expressions above, we need to compute both $\mathbb{E}_t[\Delta c_{t+1}] - \mathbb{E}_t^\mathbb{Q}[\Delta c_{t+1}]$ and $\mathbb{E}_t[w_{t+1}] - \mathbb{E}_t^\mathbb{Q}[w_{t+1}]$. Starting with $\mathbb{E}_t[\Delta c_{t+1}] - \mathbb{E}_t^\mathbb{Q}[\Delta c_{t+1}]$, we have:

$$\begin{aligned} \mathbb{E}_t[\Delta c_{t+1}] - \mathbb{E}_t^\mathbb{Q}[\Delta c_{t+1}] &= -\sigma_c \mathbb{E}_t^\mathbb{Q}[\varepsilon_{u,t+1} - \varepsilon_{d,t+1}] \\ &= \sigma_c \left(\frac{\gamma \sigma_c}{1 + \gamma \sigma_c} V_{u,t} + \frac{\gamma \sigma_c}{1 - \gamma \sigma_c} V_{d,t} \right) \\ &= \gamma \sigma_c^2 \left(\frac{1}{1 + \gamma \sigma_c} V_{u,t} + \frac{1}{1 - \gamma \sigma_c} V_{d,t} \right) \end{aligned}$$

Similarly, for $\mathbb{E}_t[w_{t+1}] - \mathbb{E}_t^\mathbb{Q}[w_{t+1}]$ we get:

$$\begin{aligned} \mathbb{E}_t[w_{t+1}] - \mathbb{E}_t^\mathbb{Q}[w_{t+1}] &= A_1 \left(\mathbb{E}_t[V_{u,t}] - \mathbb{E}_t^\mathbb{Q}[V_{u,t}] \right) + A_2 \left(\mathbb{E}_t[V_{d,t}] - \mathbb{E}_t^\mathbb{Q}[V_{d,t}] \right) \\ &\quad + A_3 \left(\mathbb{E}_t[q_{u,t}] - \mathbb{E}_t^\mathbb{Q}[q_{u,t}] \right) + A_4 \left(\mathbb{E}_t[q_{d,t}] - \mathbb{E}_t^\mathbb{Q}[q_{d,t}] \right) \end{aligned}$$

At this stage, we need the premia for each risk factor ($V_{u,t}$, $V_{d,t}$, $q_{u,t}$ and $q_{d,t}$) to compute $\mathbb{E}_t[w_{t+1}] - \mathbb{E}_t^\mathbb{Q}[w_{t+1}]$. We start with $V_{u,t}$:

$$\mathbb{E}_t[V_{u,t+1}] - \mathbb{E}_t^\mathbb{Q}[V_{u,t+1}] = -\sqrt{q_{u,t}} \mathbb{E}_t^\mathbb{Q}[z_{t+1}^u] = -\sqrt{q_{u,t}}(\theta - 1)\kappa_1 A_1 \sqrt{q_{u,t}} = -(\theta - 1)\kappa_1 A_1 q_{u,t}.$$

Thus, we derive the premia accrued to each risk factor as:

$$\begin{aligned} \mathbb{E}_t[V_{u,t+1}] - \mathbb{E}_t^\mathbb{Q}[V_{u,t+1}] &= (1 - \theta)\kappa_1 A_1 q_{u,t}, \\ \mathbb{E}_t[V_{d,t+1}] - \mathbb{E}_t^\mathbb{Q}[V_{d,t+1}] &= (1 - \theta)\kappa_1 A_2 q_{d,t}, \\ \mathbb{E}_t[q_{u,t+1}] - \mathbb{E}_t^\mathbb{Q}[q_{u,t+1}] &= (1 - \theta)\kappa_1 A_3 \varphi_u^2 q_{u,t}, \\ \mathbb{E}_t[q_{d,t+1}] - \mathbb{E}_t^\mathbb{Q}[q_{d,t+1}] &= (1 - \theta)\kappa_1 A_4 \varphi_d^2 q_{d,t}. \end{aligned}$$

Collecting these terms and substituting in the expression for $\mathbb{E}_t[w_{t+1}] - \mathbb{E}_t^\mathbb{Q}[w_{t+1}]$, we get:

$$\mathbb{E}_t[w_{t+1}] - \mathbb{E}_t^\mathbb{Q}[w_{t+1}] = (1 - \theta)\kappa_1 \left[(A_1^2 + A_3^2 \varphi_u^2) q_{u,t} + (A_2^2 + A_4^2 \varphi_d^2) q_{d,t} \right].$$

It easily follows that the equity premium in our model is:

$$ERP_t \equiv \frac{\gamma \sigma_c^2}{1 + \gamma \sigma_c} V_{u,t} + \frac{\gamma \sigma_c^2}{1 - \gamma \sigma_c} V_{d,t} + (1 - \theta)\kappa_1 (A_1^2 + A_3^2 \varphi_u^2) q_{u,t} + (1 - \theta)\kappa_1 (A_2^2 + A_4^2 \varphi_d^2) q_{d,t}. \quad (43)$$

We now characterize variance of the consumption-generating asset

$$\begin{aligned}
\sigma_{r,t}^2 &\equiv \text{Var}_t[r_{c,t+1}] \\
&= \text{Var}_t[\sigma_c(\varepsilon_{u,t+1} - \varepsilon_{d,t+1}) + \kappa_1[(A_1 z_{t+1}^u + \varphi_u A_3 z_{t+1}^1)\sqrt{q_{u,t}} + (A_2 z_{t+1}^d + \varphi_d A_4 z_{t+1}^2)\sqrt{q_{d,t}}]] \\
&= \sigma_{c,t}^2 + \kappa_1^2 \sigma_w^2 \\
&= \sigma_c^2 V_{u,t} + \sigma_c^2 V_{d,t} + \kappa_1^2 (A_1^2 + A_3^2 \varphi_u^2) q_{u,t} + \kappa_1^2 (A_2^2 + A_4^2 \varphi_d^2) q_{d,t} \\
&\equiv (\sigma_{r,t}^u)^2 + (\sigma_{r,t}^d)^2,
\end{aligned}$$

where upside and downside variances are defined as:

$$(\sigma_{r,t}^u)^2 = \sigma_c^2 V_{u,t} + \kappa_1^2 (A_1^2 + A_3^2 \varphi_u^2) q_{u,t}, \quad (44)$$

$$(\sigma_{r,t}^d)^2 = \sigma_c^2 V_{d,t} + \kappa_1^2 (A_2^2 + A_4^2 \varphi_d^2) q_{d,t}. \quad (45)$$

Using the definition of VRP , we define upside VRP as:

$$\begin{aligned}
VRP_t^u &\equiv \mathbb{E}_t^{\mathbb{Q}}[(\sigma_{r,t+1}^u)^2] - \mathbb{E}_t[(\sigma_{r,t+1}^u)^2], \\
&= \sigma_c^2(\theta - 1)\kappa_1 A_1 q_{u,t} + \kappa_1^2 (A_1^2 + A_3^2 \varphi_u^2) (\theta - 1)\kappa_1 A_3 \varphi_u^2 q_{u,t}, \\
&= (\theta - 1)(\sigma_c^2 \kappa_1 A_1 + \kappa_1^3 (A_1^2 + A_3^2 \varphi_u^2) A_3 \varphi_u^2) q_{u,t}.
\end{aligned} \quad (46)$$

Similarly, we define downside VRP as:

$$\begin{aligned}
VRP_t^d &\equiv \mathbb{E}_t^{\mathbb{Q}}[(\sigma_{r,t+1}^d)^2] - \mathbb{E}_t[(\sigma_{r,t+1}^d)^2] \\
&= (\theta - 1)(\sigma_c^2 \kappa_1 A_2 + \kappa_1^3 (A_2^2 + A_4^2 \varphi_d^2) A_4 \varphi_d^2) q_{d,t}.
\end{aligned} \quad (47)$$

Appendix A-3: Restrictions on parameters

We expect $VRP_t^U < 0$ and $VRP_t^D > 0$. It follows that

$$\sigma_c^2 \kappa_1 A_1 + \kappa_1^3 (A_1^2 + A_3^2 \varphi_u^2) A_3 \varphi_u^2 > 0, \quad (48)$$

$$\sigma_c^2 \kappa_1 A_2 + \kappa_1^3 (A_2^2 + A_4^2 \varphi_d^2) A_4 \varphi_d^2 < 0. \quad (49)$$

In this appendix, we discuss the necessary and sufficient conditions ensuring that both inequalities (48) and (49) hold.

Since

$$\begin{aligned}
A_3 &= \frac{A_3^*}{\theta}, \quad A_4 = \frac{A_4^*}{\theta}, \quad \text{with} \\
A_3^* &= \frac{(1 - \kappa_1 \gamma_{u,1}) - \sqrt{(1 - \kappa_1 \gamma_{u,1})^2 - \varphi_u^2 \kappa_1^4 (A_1^*)^2}}{\kappa_1^2 \varphi_u^2} \\
A_4^* &= \frac{(1 - \kappa_1 \gamma_{d,1}) - \sqrt{(1 - \kappa_1 \gamma_{d,1})^2 - \varphi_d^2 \kappa_1^4 (A_2^*)^2}}{\kappa_1^2 \varphi_d^2},
\end{aligned}$$

it must be that

$$\begin{aligned}
0 &\leq \varphi_u \leq \frac{1 - \kappa_1 \gamma_{u,1}}{-\kappa_1^2 A_1^*}, \\
0 &\leq \varphi_d \leq \frac{1 - \kappa_1 \gamma_{d,1}}{\kappa_1^2 A_2^*},
\end{aligned}$$

and

$$A_1^* \equiv \theta A_1 = -\frac{f(\sigma_c(1-\gamma)) + (1-\gamma)\mu_1}{\kappa_1\beta_u - 1}, \quad (50)$$

$$A_2^* \equiv \theta A_2 = -\frac{f(-\sigma_c(1-\gamma)) + (1-\gamma)\mu_2}{\kappa_1\beta_d - 1}. \quad (51)$$

Since $A_4 < 0$, $A_2 < 0$ is a sufficient condition for $\sigma_c^2\kappa_1 A_2 + \kappa_1^3 (A_2^2 + A_4^2\varphi_d^2) A_4\varphi_d^2 < 0$. Note that

$$A_2 < 0 \Leftrightarrow \mu_2 < \frac{f\left[-\sigma_c(1-\gamma)\right]}{\gamma - 1}$$

In particular, we have

$$\mu_2 \leq 0 \Rightarrow A_2 < 0 \Rightarrow VRP_t^d > 0.$$

Since $A_3 < 0$, $A_1 > 0$ is a necessary condition for $\sigma_c^2\kappa_1 A_1 + \kappa_1^3 (A_1^2 + A_3^2\varphi_u^2) A_3\varphi_u^2 > 0$.

Because we want $A_1 > 0$, it must be the case that

$$\frac{f(\sigma_c(1-\gamma))}{\gamma - 1} \leq \mu_1.$$

We can show that, the constraint $\sigma_c^2\kappa_1 A_1 + \kappa_1^3 (A_1^2 + A_3^2\varphi_u^2) A_3\varphi_u^2 > 0$ translates into

$$\theta < -\sqrt{\frac{2(\kappa_1\gamma_{u,1} - 1)}{A_1^*}} \frac{\varphi_u A_3^*}{\sigma_c}.$$

Appendix B: Linking the asymmetry to the signed jump variation

Consider that $s_t = \ln S_t$, the stock log-price evolves according to a jump-diffusion process

$$ds_t = \mu_t dt + \sigma_t dW_t + \Delta s_t, \quad (52)$$

where dW_t is an increment of standard Brownian motion and $\Delta s_t \equiv s_t - s_{t-}$ is the jump component. From equation (52), the return over a time interval $[t, t+h]$ is

$$r_{t,h} \equiv \int_t^{t+h} ds_v = \int_t^{t+h} \mu_v dv + \int_t^{t+h} \sigma_v dW_v + \sum_{t \leq v \leq t+h} \Delta s_v \quad (53)$$

The corresponding demeaned return is

$$\begin{aligned} \varepsilon_{t,h}^r &= r_{t,h} - \left(\int_t^{t+h} \mu_v dv + \mathbb{E}_t [\tilde{\varepsilon}_{t,h}^J] \right), \\ &= \varepsilon_{t,h}^C + \varepsilon_{t,h}^J, \end{aligned} \quad (54)$$

where $\tilde{\varepsilon}_{t,h}^J = \sum_{t \leq v \leq t+h} \Delta s_v$ denotes the jump component, $\varepsilon_{t,h}^J = \tilde{\varepsilon}_{t,h}^J - \mathbb{E}_t [\tilde{\varepsilon}_{t,h}^J]$ is the centered conditional jump innovation, and $\varepsilon_{t,h}^C = \int_t^{t+h} \sigma_v dW_v$ stands for the diffusion part.

Between time t and $t+h$, let $N_{t,h}$ be the total number of jumps drawn from a counting process. Thus,

the jump component in equation (54) can be rewritten as

$$\begin{aligned}
\tilde{\varepsilon}_{t,h}^J &= \sum_{j=0}^{N_{t,h}} \Delta s_j, \\
&= \sum_{j=0}^{N_{t,h}} |\Delta s_j| \mathbb{I}_{[\Delta s_j > 0]} - \sum_{j=0}^{N_{t,h}} |\Delta s_j| \mathbb{I}_{[\Delta s_j \leq 0]}, \\
&= \sum_{j=0}^{N_{t,h}^U} |\Delta s_j| - \sum_{j=0}^{N_{t,h}^D} |\Delta s_j|,
\end{aligned} \tag{55}$$

where $N_{t,h}^U = \sum_{j=0}^{N_{t,h}} \mathbb{I}_{[\Delta s_j > 0]}$ and $N_{t,h}^D = \sum_{j=0}^{N_{t,h}} \mathbb{I}_{[\Delta s_j \leq 0]}$ denote the number of upside and downside jumps, respectively. It follows that

$$\varepsilon_{t,h}^J = \varepsilon_{t,h}^U - \varepsilon_{t,h}^D, \tag{56}$$

where $\varepsilon_{t,h}^U = \sum_{j=0}^{N_{t,h}^U} |\Delta s_j| - \mathbb{E}_t \left[\sum_{j=0}^{N_{t,h}^U} |\Delta s_j| \right]$, and $\varepsilon_{t,h}^D = \sum_{j=0}^{N_{t,h}^D} |\Delta s_j| - \mathbb{E}_t \left[\sum_{j=0}^{N_{t,h}^D} |\Delta s_j| \right]$.

We further assume that absolute values of jump sizes ($|\Delta s_j|$) are independent and identically distributed according to a positive law, and that the number of upside ($N_{t,h}^U \sim \mathcal{P}(\lambda_{t,h}^U)$) and downside ($N_{t,h}^D \sim \mathcal{P}(\lambda_{t,h}^D)$) jumps are independent Poisson distributed random variables with distinct intensities.

In the sequel, we drop time subscripts to ease notation. We now compute the first three cumulants of ε^J . The cumulant generating function is

$$\psi_{\varepsilon^J}(\nu) = \psi_{\varepsilon^U}(\nu) + \psi_{\varepsilon^D}(-\nu). \tag{57}$$

Thus, the k^{th} derivative of $\psi_{\varepsilon^J}(\nu)$ evaluated at 0 yields the k^{th} cumulant μ_k . We get the following equalities

$$\mu_1(\varepsilon^J) = \mu_1(\varepsilon^U) - \mu_1(\varepsilon^D) = 0, \tag{58}$$

$$\mu_2(\varepsilon^J) = \mu_2(\varepsilon^U) + \mu_2(\varepsilon^D), \tag{59}$$

$$\mu_3(\varepsilon^J) = \mu_3(\varepsilon^U) - \mu_3(\varepsilon^D). \tag{60}$$

Moreover, the distributional assumptions imply

$$\mu_2(\varepsilon^{U/D}) = \lambda^{U/D} (\bar{\sigma}^2 + \bar{m}^2), \tag{61}$$

and

$$\mu_3(\varepsilon^{U/D}) = \lambda^{U/D} (\bar{s}k^3 + 3\bar{\sigma}^2\bar{m} + \bar{m}^3), \tag{62}$$

where $\bar{m} = \mu_1(|\Delta s_j|)$, $\bar{\sigma}^2 = \mu_2(|\Delta s_j|)$, and $\bar{s}k = \mu_3(|\Delta s_j|)$.

Combining equations (60) and (62), we obtain

$$\mu_3(\varepsilon^J) = (\lambda^U - \lambda^D) (\bar{s}k^3 + 3\bar{\sigma}^2\bar{m} + \bar{m}^3). \tag{63}$$

Interestingly, we see from equation (61) that the difference between variances of ε^U and ε^D is,

$$\mu_2(\varepsilon^U) - \mu_2(\varepsilon^D) = (\lambda^U - \lambda^D) (\bar{\sigma}^2 + \bar{m}^2), \tag{64}$$

which implies that

$$(\lambda^U - \lambda^D) = (\mu_2(\varepsilon^U) - \mu_2(\varepsilon^D)) (\bar{\sigma}^2 + \bar{m}^2)^{-1}. \quad (65)$$

Substituting equation (65) into equation (60) gives

$$\mu_3(\varepsilon^J) = (\mu_2(\varepsilon^U) - \mu_2(\varepsilon^D)) (\bar{\sigma}^2 + \bar{m}^2)^{-1} (\bar{\sigma}^3 + 3\bar{\sigma}^2\bar{m} + \bar{m}^3), \quad (66)$$

which clearly shows that the difference between the variances of ε^U and ε^D drives the asymmetry. Finally, when diffusion and jump components are independent, the skewness of the return is

$$\mu_3(r) \equiv \mu_3(\varepsilon^r) = \mu_3(\varepsilon^C) + \mu_3(\varepsilon^J), \quad (67)$$

which sums the skewness of diffusive ($\mu_3(\varepsilon^C)$) and jump ($\mu_3(\varepsilon^J)$) innovations. In the absence of leverage effect, $\mu_3(\varepsilon^C) = 0$ inducing that the total of returns boils down (up to a constant) to the difference of ε^U and ε^D variances. Obviously, the larger the leverage effect, the wider the wedge between the return skewness and the difference of ε^U and ε^D variances.

Table 1: **Summary Statistics**

	Mean (%)	Median (%)	Std. Dev. (%)	Skewness	Kurtosis	AR(1)
Panel A: Excess Returns						
Equity	1.9771	14.5157	20.9463	-0.1531	10.5559	-0.0819
Equity (1996-2007)	3.0724	12.5824	17.6474	-0.1379	5.9656	-0.0165
Panel B: Risk-Neutral						
Variance	19.3544	18.7174	6.6110	1.5650	7.6100	0.9466
Downside Variance	16.9766	16.2104	5.8727	1.6746	8.0637	0.9548
Upside Variance	9.2570	9.1825	3.1295	1.1479	6.0030	0.8991
Skewness	-7.7196	-7.0090	3.0039	-2.0380	9.6242	0.7323
Panel C: Realized						
Variance	16.7137	15.3429	5.5216	3.6748	25.6985	0.9667
Downside Variance	11.7677	10.8670	3.9857	3.9042	29.4323	0.9603
Upside Variance	11.8550	10.8683	3.8639	3.6288	25.3706	0.9609
Skewness	0.0872	0.1315	1.0911	-6.3619	170.4998	0.6319
Panel D: Risk Premium						
Variance	2.6407	2.3932	4.2538	-0.3083	6.8325	0.9265
Downside Variance	5.2089	4.8693	3.8159	0.2019	4.6310	0.9444
Upside Variance	-2.5979	-2.5730	2.5876	-2.2178	22.7198	0.8877
Skewness	-7.8068	-6.9942	3.0606	-2.0696	10.6270	0.9345
Panel E: Macroeconomic						
Consumption Growth	1.4733	1.4443	1.2808	0.6384	5.1933	-0.2915
Real Interest Rate	1.0707	0.9607	1.9070	0.7341	5.0354	0.9878

This table reports the summary statistics for the quantities investigated in this study. Mean, median, and standard deviation values are annualized and in percentages. We report excess kurtosis values. $AR(1)$ represents the values for the first autocorrelation coefficient. The full sample is from September 1996 to March 2015. We also consider a sub-sample ending in December 2007. The data for seasonally adjusted, real per capita consumption growth (based on 2009 chained prices) starts in February 1999 and ends in March 2015. We report annualized mean, median, and standard deviation values for consumption growth and real interest rates.

Table 2: **Predictive Content of Premium Measure**

h	1		3		6		12	
	t -Stat	$\bar{R}^2$	t -Stat	$\bar{R}^2$	t -Stat	$\bar{R}^2$	t -Stat	$\bar{R}^2$
k	Panel A: Variance Risk Premium							
1	2.43	2.61	2.51	2.83	1.02	0.02	0.68	-0.30
2	2.84	3.76	3.42	5.58	1.50	0.68	1.04	0.05
3	4.11	8.13	3.58	6.18	1.78	1.19	1.56	0.78
6	2.78	3.65	2.24	2.22	1.57	0.82	2.09	1.87
9	1.98	1.65	1.94	1.57	1.47	0.66	1.98	1.64
12	1.96	1.64	1.43	0.61	1.53	0.77	1.73	1.14
k	Panel B: Downside Variance Risk Premium							
1	2.57	2.99	2.68	3.30	1.27	0.34	0.95	-0.06
2	3.22	4.92	4.08	7.95	2.07	1.78	1.54	0.74
3	4.76	10.72	4.46	9.50	2.61	3.12	2.32	2.37
6	3.72	6.75	3.42	5.70	2.84	3.85	3.21	4.98
9	2.96	4.27	3.14	4.86	2.82	3.86	2.99	4.35
12	3.04	4.60	2.65	3.39	2.81	3.86	2.80	3.84
k	Panel C: Upside Variance Risk Premium							
1	2.08	1.79	1.91	1.44	0.44	-0.44	-0.04	-0.55
2	2.15	1.96	2.15	1.96	0.39	-0.47	-0.18	-0.54
3	3.05	4.41	2.07	1.79	0.26	-0.52	-0.27	-0.52
6	1.57	0.82	0.61	-0.36	-0.40	-0.48	-0.27	-0.53
9	0.83	-0.18	0.36	-0.50	-0.52	-0.42	-0.14	-0.57
12	0.74	-0.27	-0.05	-0.59	-0.33	-0.52	-0.41	-0.49
k	Panel D: Skewness Risk Premium							
1	-0.10	-0.55	0.41	-0.46	0.96	-0.04	1.25	0.30
2	0.61	-0.35	1.67	0.98	1.98	1.59	2.16	1.98
3	1.03	0.04	2.24	2.18	2.81	3.70	3.29	5.17
6	2.27	2.30	3.33	5.38	4.05	8.00	4.45	9.59
9	2.57	3.13	3.39	5.70	4.20	8.73	3.98	10.59
12	2.83	3.95	3.43	5.93	3.88	7.60	4.07	8.34

This table reports predictive regression results for prediction horizons (k) between 1 and 12 months ahead, and aggregation levels (h) between 1 and 12 months, based on a predictive regression model of the form $r_{t \rightarrow t+k} = \beta_0 + \beta_1 x_t(h) + \varepsilon_{t \rightarrow t+k}$. In this regression model, $r_{t \rightarrow t+k}$ is the cumulative excess returns between t and $t+k$; $x_t(h)$ is the proposed variance or skewness risk premia component that takes the values from variance risk, upside variance risk, downside variance risk, or skewness risk premia measures; and $\varepsilon_{t \rightarrow t+k}$ is a zero-mean error term. The reported Student's t -statistics for slope parameters are constructed from heteroscedasticity and serial correlation consistent standard errors that explicitly take account of the overlap in the regressions, following Hodrick (1992). $\bar{R}^2$ represents adjusted R^2 s.

Table 3: **Predictive Content of Risk-Neutral Measure**

h	1		3		6		12	
	t -Stat	$\bar{R}^2$	t -Stat	$\bar{R}^2$	t -Stat	$\bar{R}^2$	t -Stat	$\bar{R}^2$
k	Panel A: Risk-Neutral Variance							
1	0.28	-0.51	0.50	-0.41	0.69	-0.29	0.75	-0.24
2	1.14	0.17	1.24	0.30	1.35	0.44	1.39	0.51
3	1.30	0.38	1.52	0.72	1.83	1.28	2.13	1.92
6	2.10	1.88	2.33	2.43	2.76	3.57	3.21	4.95
9	2.32	2.44	2.55	3.05	2.95	4.22	3.15	4.85
12	2.21	2.20	2.45	2.82	2.89	4.11	3.30	5.45
k	Panel B: Risk-Neutral Downside Variance							
1	0.27	-0.51	0.57	-0.37	0.77	-0.23	0.87	-0.14
2	1.22	0.27	1.39	0.51	1.49	0.66	1.54	0.74
3	1.42	0.56	1.70	1.04	2.03	1.70	2.35	2.44
6	2.23	2.17	2.52	2.91	2.97	4.21	3.42	5.67
9	2.43	2.71	2.68	3.42	3.10	4.70	3.26	5.22
12	2.32	2.48	2.55	3.09	2.99	4.42	3.42	5.84
k	Panel C: Risk-Neutral Upside Variance							
1	0.29	-0.50	0.27	-0.51	0.36	-0.48	0.20	-0.53
2	0.93	-0.07	0.76	-0.23	0.78	-0.21	0.60	-0.35
3	0.99	-0.02	0.94	-0.06	1.04	0.05	1.00	0.00
6	1.74	1.12	1.72	1.09	1.92	1.48	2.05	1.77
9	2.01	1.71	2.09	1.89	2.31	2.42	2.38	2.59
12	1.90	1.49	2.09	1.92	2.45	2.83	2.57	3.17
k	Panel D: Risk-Neutral Skewness							
1	0.22	-0.52	0.87	-0.14	1.10	0.11	1.27	0.34
2	1.51	0.70	2.02	1.67	2.06	1.76	2.08	1.79
3	1.93	1.48	2.47	2.74	2.85	3.80	3.13	4.65
6	2.70	3.42	3.28	5.21	3.80	7.02	4.11	8.18
9	2.76	3.64	3.17	4.92	3.64	6.54	3.56	6.26
12	2.67	3.44	2.88	4.07	3.27	5.35	3.66	6.73

This table reports predictive regression results for risk-neutral variance and skewness measures. The predictive regression model, prediction horizons, aggregation levels, and notation are the same as in the results reported in Table 2. The difference is in the definition of $x_t(h)$: Instead of risk premia, we use risk-neutral measures for variance, upside variance, downside variance, and skewness. The reported Student's t -statistics for slope parameters are constructed from heteroscedasticity and serial correlation consistent standard errors that explicitly take account of the overlap in the regressions, following Hodrick (1992). $\bar{R}^2$ represents adjusted R^2 s.

Table 4: **Predictive Content of Realized (Physical) Measure**

h	1		3		6		12	
	t -Stat	$\bar{R}^2$	t -Stat	$\bar{R}^2$	t -Stat	$\bar{R}^2$	t -Stat	$\bar{R}^2$
k	Panel A: Realized Variance							
1	-1.10	0.12	-0.99	-0.01	-0.10	-0.55	0.09	-0.55
2	-0.67	-0.30	-0.86	-0.15	0.18	-0.54	0.40	-0.47
3	-1.18	0.22	-0.75	-0.25	0.36	-0.48	0.62	-0.34
6	0.01	-0.57	0.48	-0.44	1.13	0.15	1.07	0.09
9	0.55	-0.40	0.78	-0.22	1.33	0.44	1.12	0.15
12	0.49	-0.45	0.98	-0.02	1.27	0.35	1.40	0.56
k	Panel B: Realized Downside Variance							
1	-1.05	0.06	-0.90	-0.10	-0.08	-0.55	0.09	-0.55
2	-0.53	-0.40	-0.76	-0.23	0.21	-0.53	0.39	-0.47
3	-1.04	0.05	-0.68	-0.30	0.39	-0.48	0.59	-0.36
6	0.05	-0.57	0.48	-0.44	1.08	0.10	0.99	-0.01
9	0.54	-0.41	0.75	-0.25	1.21	0.27	1.01	0.01
12	0.44	-0.48	0.89	-0.12	1.14	0.18	1.30	0.40
k	Panel C: Realized Upside Variance							
1	-1.15	0.18	-1.09	0.10	-0.13	-0.54	0.10	-0.55
2	-0.82	-0.18	-0.95	-0.05	0.14	-0.54	0.41	-0.46
3	-1.33	0.43	-0.82	-0.18	0.34	-0.49	0.66	-0.32
6	-0.05	-0.57	0.48	-0.44	1.17	0.21	1.16	0.19
9	0.39	-0.41	0.81	-0.19	1.44	0.61	1.23	0.29
12	0.54	-0.42	1.08	0.09	1.39	0.54	1.51	0.74
k	Panel D: Realized Skewness							
1	0.44	-0.45	1.58	0.81	0.63	-0.33	-0.06	-0.55
2	1.51	0.70	1.67	0.99	0.96	-0.05	-0.26	-0.52
3	1.45	0.61	1.19	0.23	0.71	-0.28	-0.89	-0.12
6	0.54	-0.40	0.11	-0.56	-1.01	0.01	-2.64	3.25
9	0.07	-0.58	-0.54	-0.41	-3.01	4.41	-3.67	6.67
12	-0.55	-0.41	-1.82	1.33	-3.37	5.70	-3.36	5.67

This table reports predictive regression results for realized variance and skewness measures. The predictive regression model, prediction horizons, aggregation levels, and notation are the same as in the results reported in Table 2. The difference is in the definition of $x_t(h)$: Instead of risk premia, we use realized (historical) measures for variance, upside variance, downside variance, and skewness. The reported Student's t -statistics for slope parameters are constructed from heteroscedasticity and serial correlation consistent standard errors that explicitly take account of the overlap in the regressions, following Hodrick (1992). $\bar{R}^2$ represents adjusted R^2 s.

Table 5: Joint Regression Results

h	1			3			6			12		
	t -Stat		$\bar{R}^2$	t -Stat		$\bar{R}^2$	t -Stat		$\bar{R}^2$	t -Stat		$\bar{R}^2$
k	Up	Down		Up	Down		Up	Down		Up	Down	
Panel A: Risk Premium												
1	-0.01	1.49	2.45	-0.12	1.86	2.77	-0.58	1.32	-0.03	-0.85	1.27	-0.21
2	-0.78	2.49	4.72	-1.28	3.66	8.28	-1.38	2.46	2.26	-1.54	2.17	1.49
3	-1.28	3.79	11.04	-1.81	4.32	10.63	-2.06	3.33	4.84	-2.34	3.31	4.77
6	-2.46	4.19	9.36	-3.00	4.56	9.80	-3.31	4.39	8.98	-3.14	4.54	9.54
9	-2.75	3.99	7.76	-3.10	4.45	9.36	-3.46	4.49	9.50	-2.74	4.09	7.83
12	-3.06	4.29	9.06	-3.18	4.18	8.30	-3.13	4.24	8.61	-2.97	4.10	8.06
Panel B: Risk-Neutral Measures												
1	0.08	-0.01	-1.06	-1.13	1.24	-0.22	-1.30	1.46	0.15	-1.48	1.70	0.51
2	-1.00	1.27	0.27	-2.42	2.69	3.10	-2.27	2.61	2.90	-2.00	2.45	2.35
3	-1.59	1.89	1.39	-2.92	3.26	5.01	-3.22	3.68	6.55	-2.87	3.59	6.21
6	-1.64	2.14	3.09	-2.93	3.47	6.89	-3.25	3.99	9.12	-2.61	3.79	8.66
9	-1.32	1.88	3.12	-2.02	2.62	5.09	-2.25	3.05	6.88	-1.42	2.62	5.78
12	-1.35	1.89	2.94	-1.49	2.07	3.77	-1.39	2.18	4.93	-1.28	2.55	6.20
Panel C: Realized (Physical) Measures												
1	-0.63	0.42	-0.28	-1.82	1.72	1.17	-0.69	0.68	-0.84	0.11	-0.10	-1.10
2	-1.62	1.50	0.51	-1.92	1.83	1.24	-0.93	0.95	-0.60	0.48	-0.46	-0.90
3	-1.68	1.46	1.05	-1.41	1.34	0.25	-0.62	0.64	-0.82	1.27	-1.24	-0.02
6	-0.54	0.54	-0.97	-0.01	0.06	-1.01	1.39	-1.31	0.61	3.54	-3.49	6.14
9	0.01	0.08	-0.99	0.72	-0.64	-0.54	3.61	-3.52	6.77	4.82	-4.76	11.39
12	0.63	-0.55	-0.83	2.07	-1.98	1.77	3.99	-3.91	8.24	4.66	-4.59	11.22

This table reports predictive regression results when multiple variance components (risk premia, risk-neutral, and realized measures) are included in the regression model. The prediction horizons, aggregation levels, and notation are the same as in the results reported in Table 2. The difference is in the regression model. Both upside and downside variance components are in the model: $r_{t \rightarrow t+k} = \beta_0 + \beta_1 x_{1,t}(h) + \beta_2 x_{2,t}(h) + \varepsilon_{t \rightarrow t+k}$. $x_{1,t}(h)$ pertains to upside measures and $x_{2,t}(h)$ represents the downside measures used in the analysis. The reported Student's t -statistics for slope parameters are constructed from heteroscedasticity and serial correlation consistent standard errors that explicitly take account of the overlap in the regressions, following Hodrick (1992). $\bar{R}^2$ represents adjusted R^2 s.

Table 6: Semi-Annual Simple Predictive Regressions, September 1996 to March 2015

<i>Intercept</i>	0.0005 (0.1995)	-0.0026 (-1.1632)	-0.0132 (-3.0304)	0.0012 (0.7230)	0.0756 (2.9076)	0.0861 (3.1978)	0.0514 (2.0974)	0.0004 (0.1293)	-0.0000 (-0.0067)	0.0210 (6.3627)	-0.0105 (-3.1169)	-0.0093 (-2.0903)
<i>uwrp_t</i>	-0.0179 (-0.3917)											
<i>dvrp_t</i>		0.1135 (2.5900)										
<i>srp_t</i>			-0.1838 (-3.6007)									
<i>vrp_t</i>				0.0516 (1.4937)								
<i>log(p_t/d_t)</i>					-0.0419 (-2.8630)							
<i>log(p_{t-1}/d_t)</i>						-0.0478 (-3.1550)						
<i>log(p_t/e_t)</i>							-0.0384 (-2.0487)					
<i>tmst_t</i>								0.7181 (0.4233)				
<i>dfst_t</i>									1.6755 (0.3885)			
<i>inflt_t</i>										-0.8052 (-6.7137)		
<i>kpi_t</i>											0.1472 (4.0170)	
<i>kpost_t</i>												0.1203 (2.5798)
<i>Adj. R² (%)</i>	-0.5157	3.3439	6.7611	0.7406	4.1793	5.1472	1.9009	-0.5000	-0.5172	21.0807	8.4028	3.3140

This table presents predictive regressions of the semi-annually (scaled) cumulative excess return $r_{t \rightarrow t+6}^e / 6 = \sum_{j=1}^6 r_{t+j}^e / 6$ on each one-period (1-month) lagged predictor from September 1996 to March 2015. The Student's t-statistics presented in parentheses below the estimated coefficients are constructed from heteroscedasticity and serial correlation consistent standard errors that explicitly take account of the overlap in the regressions, following Hodrick (1992).

Table 7: Semi-Annual Multiple Predictive Regressions, September 1996 to March 2015

<i>Intercept</i>	-0.0128 (-2.9375)	-0.0128 (-2.9375)	-0.0133 (-2.9489)	0.0787 (3.0953)	0.0951 (3.6144)	0.0582 (2.4206)	-0.0045 (-1.3553)	-0.0084 (-1.7752)	0.0171 (4.9989)	-0.0137 (-3.8625)	-0.0131 (-2.8590)
<i>wvrp_t</i>	-0.1545 (-2.7123)										
<i>dvrp_t</i>	0.2100 (3.7629)	0.0555 (1.1549)	0.4321 (3.4605)	0.1272 (2.9688)	0.1392 (3.2550)	0.1309 (2.9980)	0.1178 (2.6638)	0.1359 (2.9172)	0.1289 (3.3499)	0.1048 (2.4918)	0.1129 (2.6228)
<i>srp_t</i>		-0.1545 (-2.7123)									
<i>vrp_t</i>			-0.2640 (-2.7177)								
<i>log(p_t/d_t)</i>				-0.0462 (-3.2112)							
<i>log(p_{t-1}/d_t)</i>					-0.0557 (-3.7277)						
<i>log(p_t/e_t)</i>						-0.0471 (-2.5413)					
<i>tmst</i>							1.2900 (0.7681)				
<i>dfs_t</i>								6.2346 (1.3864)			
<i>infl_t</i>									-0.8272 (-7.0977)		
<i>kpis_t</i>										0.1425 (3.9439)	
<i>kpos_t</i>											0.1197 (2.6128)
<i>Adj. R² (%)</i>	6.9505	6.9505	6.9664	8.5372	10.3900	6.4572	3.1016	3.8843	25.7111	11.2225	6.6602

This table presents predictive regressions of the semi-annually (scaled) cumulative excess return $r_{t \rightarrow t+6}^e / 6 = \sum_{i=1}^6 r_{t+i}^e / 6$ on one-period (1-month) lagged downside *VRP* *dvrp* and one alternative predictor in turn from September 1996 to March 2015. The Student's t-statistics presented in parentheses below the estimated coefficients are constructed from heteroscedasticity and serial correlation consistent standard errors that explicitly take account of the overlap in the regressions, following Hodrick (1992).

Table 8: Semi-Annual Simple Predictive Regressions, September 1996 to December 2007

<i>Intercept</i>	0.0129 (5.1388)	-0.0070 (-3.5015)	-0.0124 (-2.6470)	0.0014 (1.0794)	0.2016 (6.6115)	0.2164 (7.0123)	0.0913 (3.9851)	0.0035 (1.5568)	0.0206 (3.4844)	0.0052 (1.0608)	-0.0059 (-2.2330)	-0.0047 (-1.3202)
<i>uwrp_t</i>	0.2679 (4.5558)											
<i>dvrp_t</i>		0.2784 (6.6863)										
<i>srp_t</i>			-0.2156 (-3.5176)									
<i>vrp_t</i>				0.2300 (6.5106)								
<i>log(p_t/d_t)</i>					-0.1092 (-6.5085)							
<i>log(p_{t-1}/d_t)</i>						-0.1176 (-6.9106)						
<i>log(p_t/e_t)</i>							-0.0662 (-3.8474)					
<i>tms_t</i>								-0.1990 (-0.1295)				
<i>dfs_t</i>									-25.6836 (-3.0118)			
<i>inflt_t</i>										-0.0735 (-0.4029)		
<i>kpis_t</i>											0.1217 (4.0782)	
<i>kpost_t</i>												0.0940 (2.4679)
<i>Adj. R² (%)</i>	13.2802	25.3069	8.1025	24.2902	24.2781	26.6030	9.6656	-0.7680	5.8882	-0.6536	10.8080	3.7964

This table presents predictive regressions of the semi-annually (scaled) cumulative excess return $r_{t \rightarrow t+6}^e = \sum_{j=1}^6 r_{t+j}^e$ on each one-period (1-month) lagged predictor from September 1996 to December 2007. The Student's t-statistics presented in parentheses below the estimated coefficients are constructed from heteroscedasticity and serial correlation consistent standard errors that explicitly take account of the overlap in the regressions, following Hodrick (1992).

Table 9: Semi-Annual Multiple Predictive Regressions, September 1996 to December 2007

<i>Intercept</i>	-0.0045 (-1.0099)	-0.0045 (-1.0099)	-0.0049 (-1.0492)	0.1736 (6.6158)	0.1954 (7.5813)	0.1053 (5.7055)	-0.0087 (-3.2688)	-0.0033 (-0.4896)	-0.0142 (-2.8139)	-0.0155 (-5.8453)	-0.0162 (-4.7195)
<i>wvrp_t</i>	0.0454 (0.6200)										
<i>dvrp_t</i>	0.2554 (4.5711)	0.3008 (5.4521)	0.2065 (1.4127)	0.2534 (7.0696)	0.2629 (7.6580)	0.3156 (8.4728)	0.2853 (6.7545)	0.2669 (5.7833)	0.2980 (6.8848)	0.2716 (6.9931)	0.2847 (7.0767)
<i>srp_t</i>		0.0454 (0.6200)									
<i>vrp_t</i>			0.0632 (0.5133)								
<i>log(p_t/d_t)</i>			-0.0990 (-6.8974)								
<i>log(p_{t-1}/d_t)</i>				-0.1113 (-7.8692)							
<i>log(p_t/e_t)</i>					-0.0855 (-6.1130)						
<i>tmst_t</i>							1.3106 (0.9772)				
<i>dfs_t</i>								-4.9164 (-0.5838)			
<i>infl_t</i>									0.2541 (1.5553)		
<i>kpist_t</i>										0.1148 (4.5066)	
<i>kpos_t</i>											0.1050 (3.2382)
<i>Adj. R²(%)</i>	24.9460	24.9460	24.8746	45.2343	49.3937	41.8336	25.2805	24.9203	26.1258	35.0975	30.4605

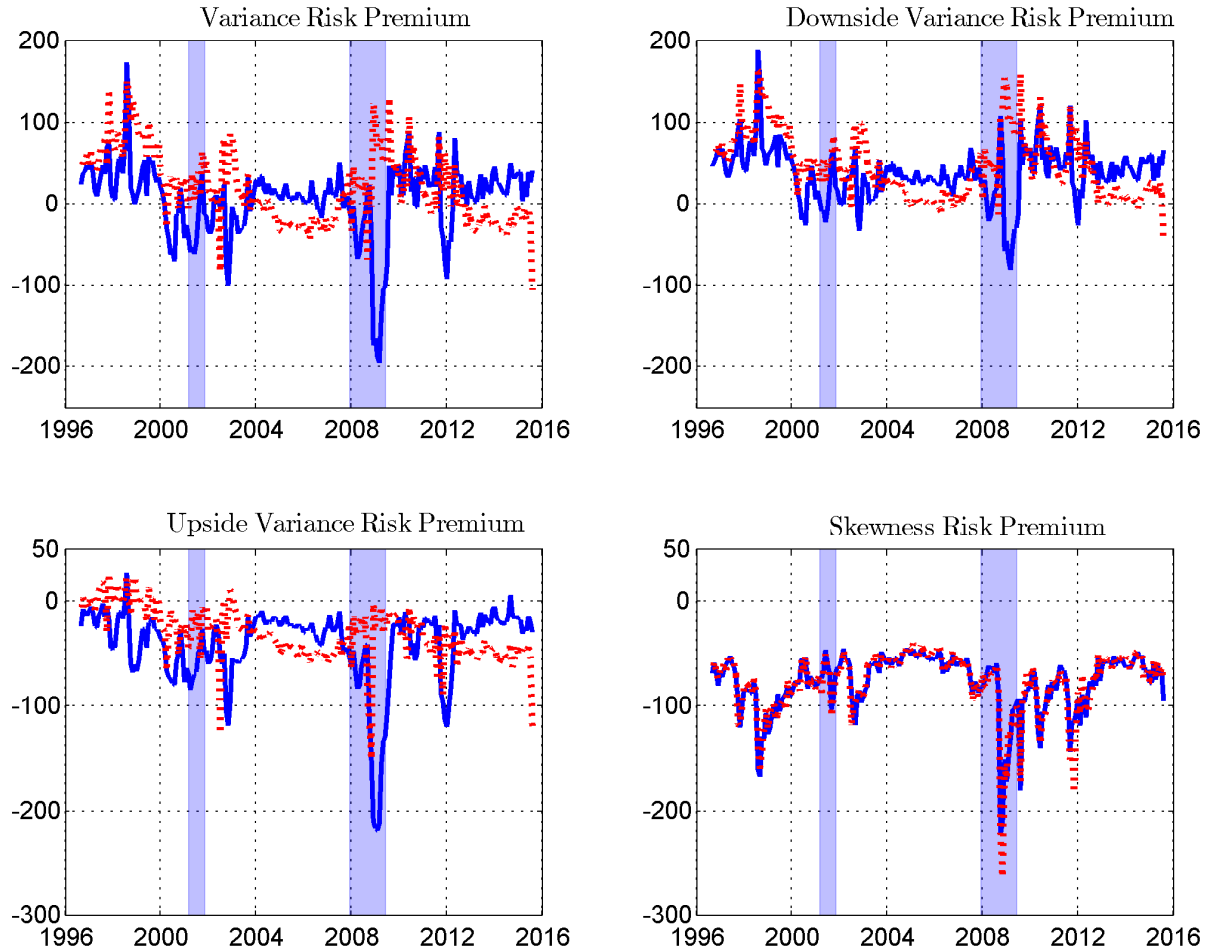
This table presents predictive regressions of the semi-annually (scaled) cumulative excess return $r_{t \rightarrow t+6}^e / 6 = \sum_{j=1}^6 r_{t+j}^e / 6$ on one-period (1-month) lagged downside *VRP* *dvrp* and one alternative predictor in turn from September 1996 to December 2007. The Student's *t*-statistics presented in parentheses below the estimated coefficients are constructed from heteroscedasticity and serial correlation consistent standard errors that explicitly take account of the overlap in the regressions, following Hodrick (1992).

Table 10: **Structural Estimation of the Theoretical Model**

Parameters	Estimates	Std. Err.
γ	1.01	0.00
θ	-0.04	0.02
δ	1.00	0.00
μ_0	0.32	0.10
μ_1	0.05	0.01
μ_2	-0.10	0.01
σ_c	0.87	0.04
α_u	0.30	0.03
β_u	0.99	0.00
α_d	0.16	0.01
β_d	0.99	0.00
$\gamma_{u,0}$	8.22E+03	3.07E+03
$\gamma_{u,1}$	0.53	0.00
φ_u	1.14E+05	4.72E+04
$\gamma_{d,0}$	1.27E+03	2.01E+02
$\gamma_{d,1}$	0.55	0.00
φ_d	4.76E+04	1.75E+04
κ_0	0.00	0.00
κ_1	0.13	0.04

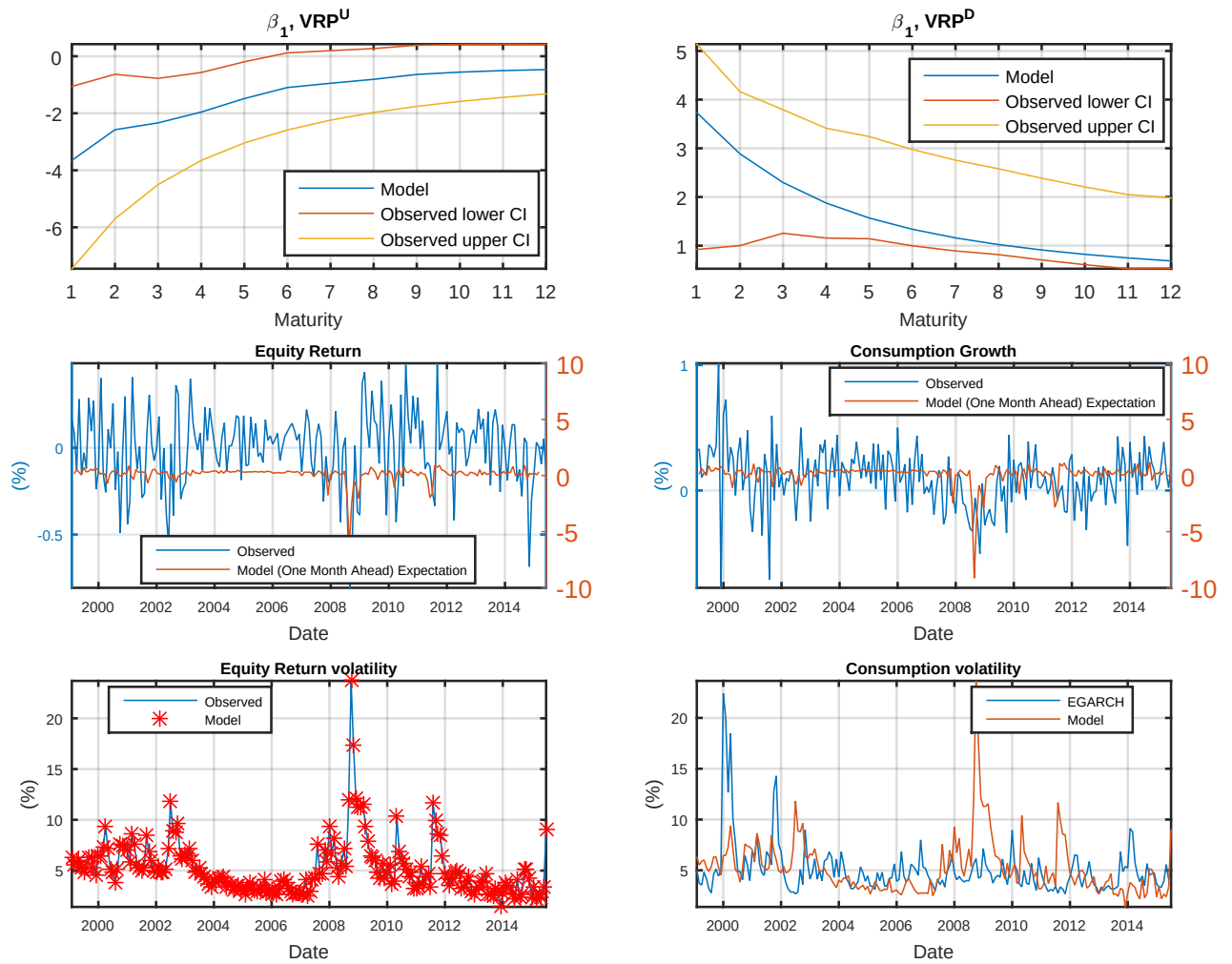
This table reports the structural parameter estimates of the general equilibrium model along with their standard errors (Std. Err.). These estimates are obtained by maximizing the joint likelihood of consumption growth, stock return and risk-free rate.

Figure 1: Time Series for Variance and Skewness Risk Premia



These figures plot the paths of annualized monthly values ($\times 10^3$) for the *VRP*, upside *VRP*, downside *VRP*, and skewness risk premium, extracted from U.S. financial markets data for September 1996 to March 2015. Solid lines represent premia constructed from random walk forecasts of the realized volatility and components. The dotted lines represent the same for M-HAR forecasts of the realized volatility and components. The shaded areas represent NBER recessions.

Figure 2: Confronting the General Equilibrium Model with Data



The top plots show the slopes of the regressions (as per equation (10)) of excess return on upside (resp. downside) VRP implied by the general equilibrium model at different maturities (expressed in months), along with the observed 95% confidence intervals. The middle (resp. bottom) plots present monthly model-implied equity return and consumption growth (resp. volatility) paths against the observed corresponding series.